

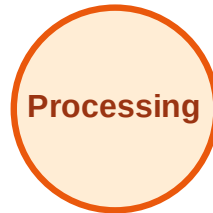
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing

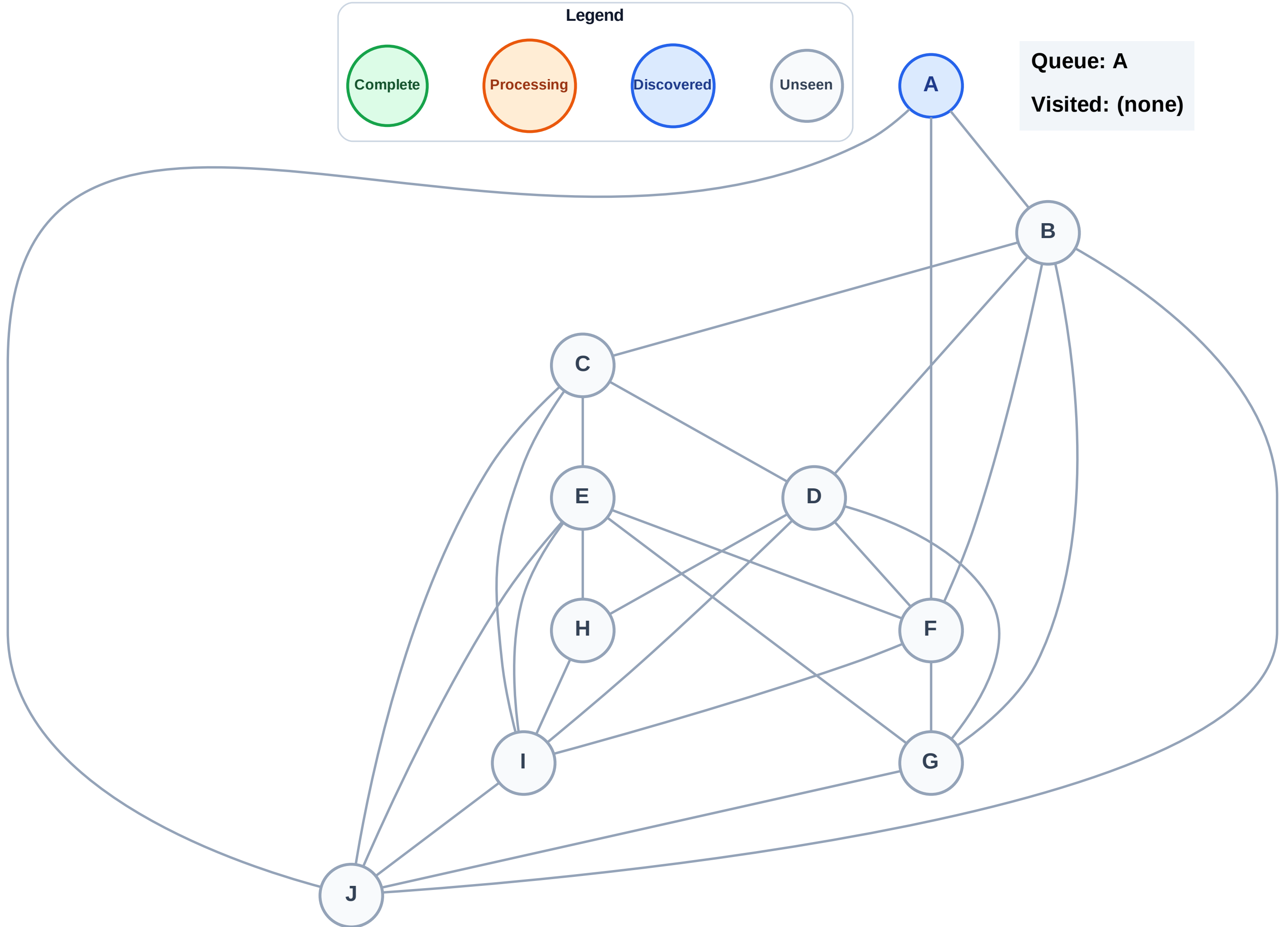


Discovered



Unseen

Queue: A
Visited: (none)



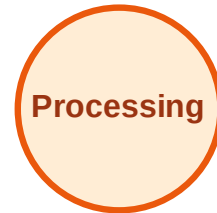
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



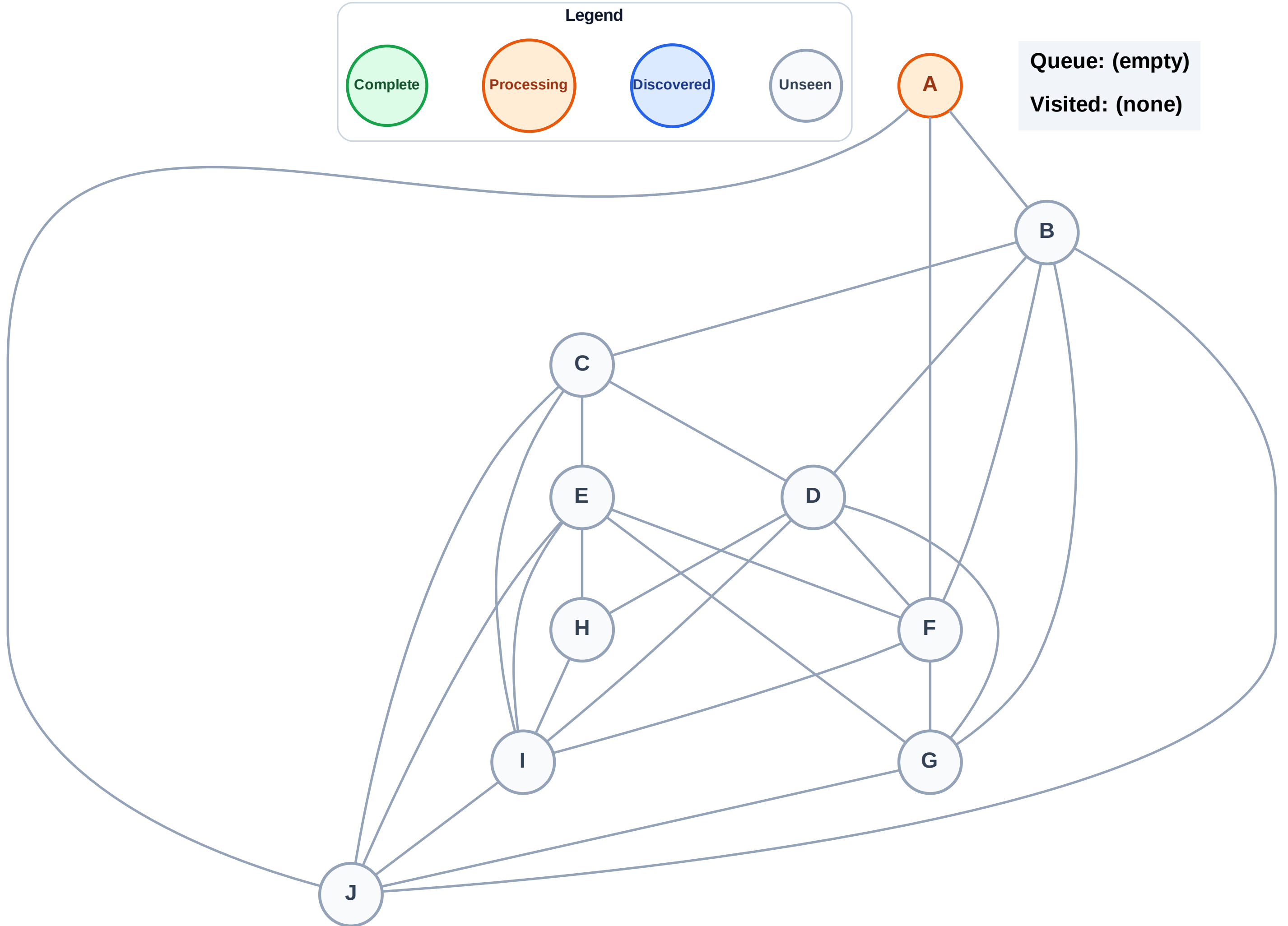
Discovered



Unseen

Queue: (empty)

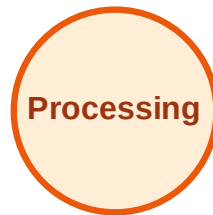
Visited: (none)



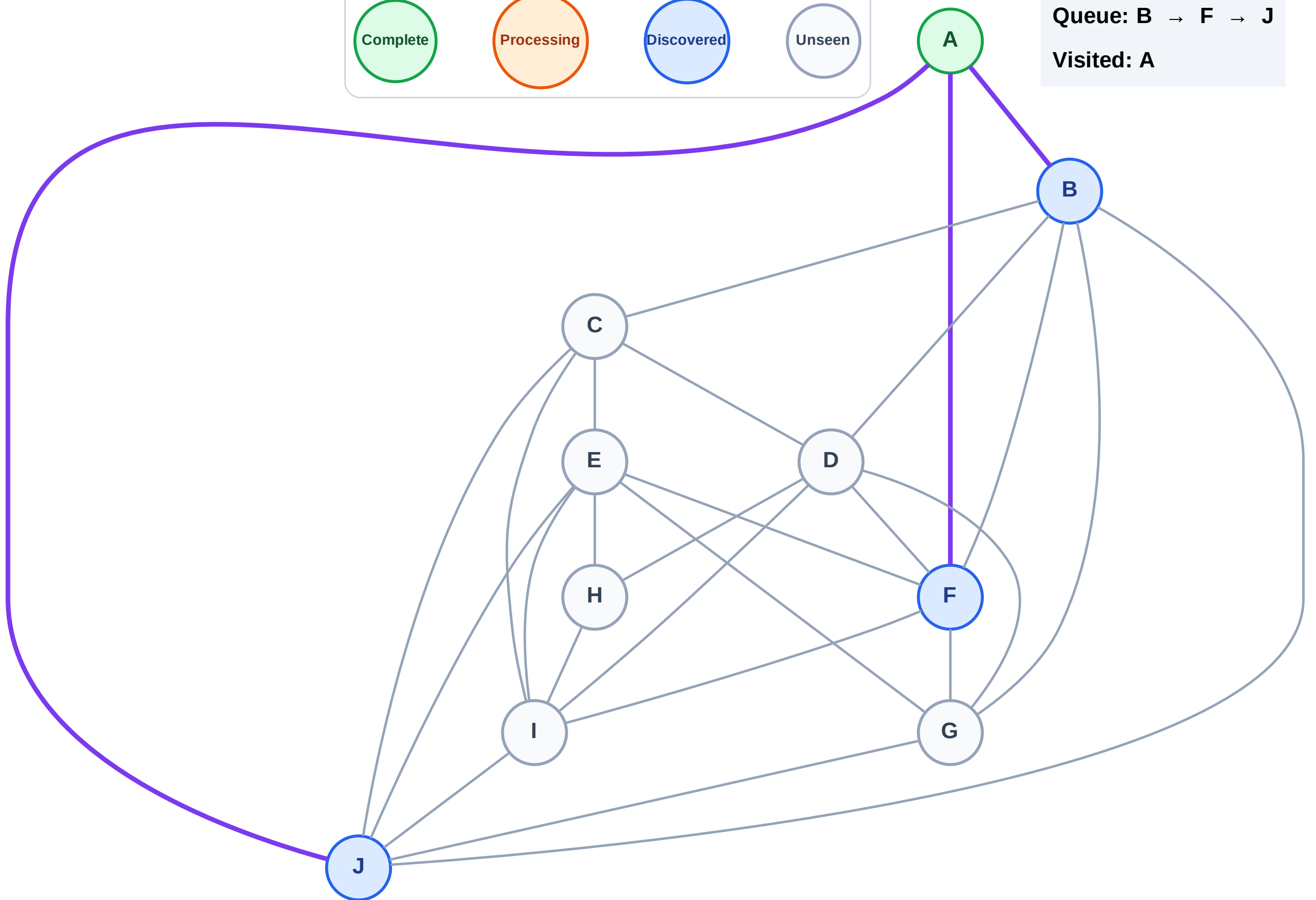
BFS Traversal

Step 3: Discover B, F, J from A

Legend



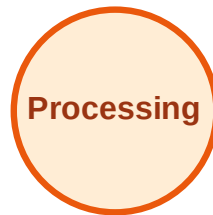
Queue: B → F → J
Visited: A



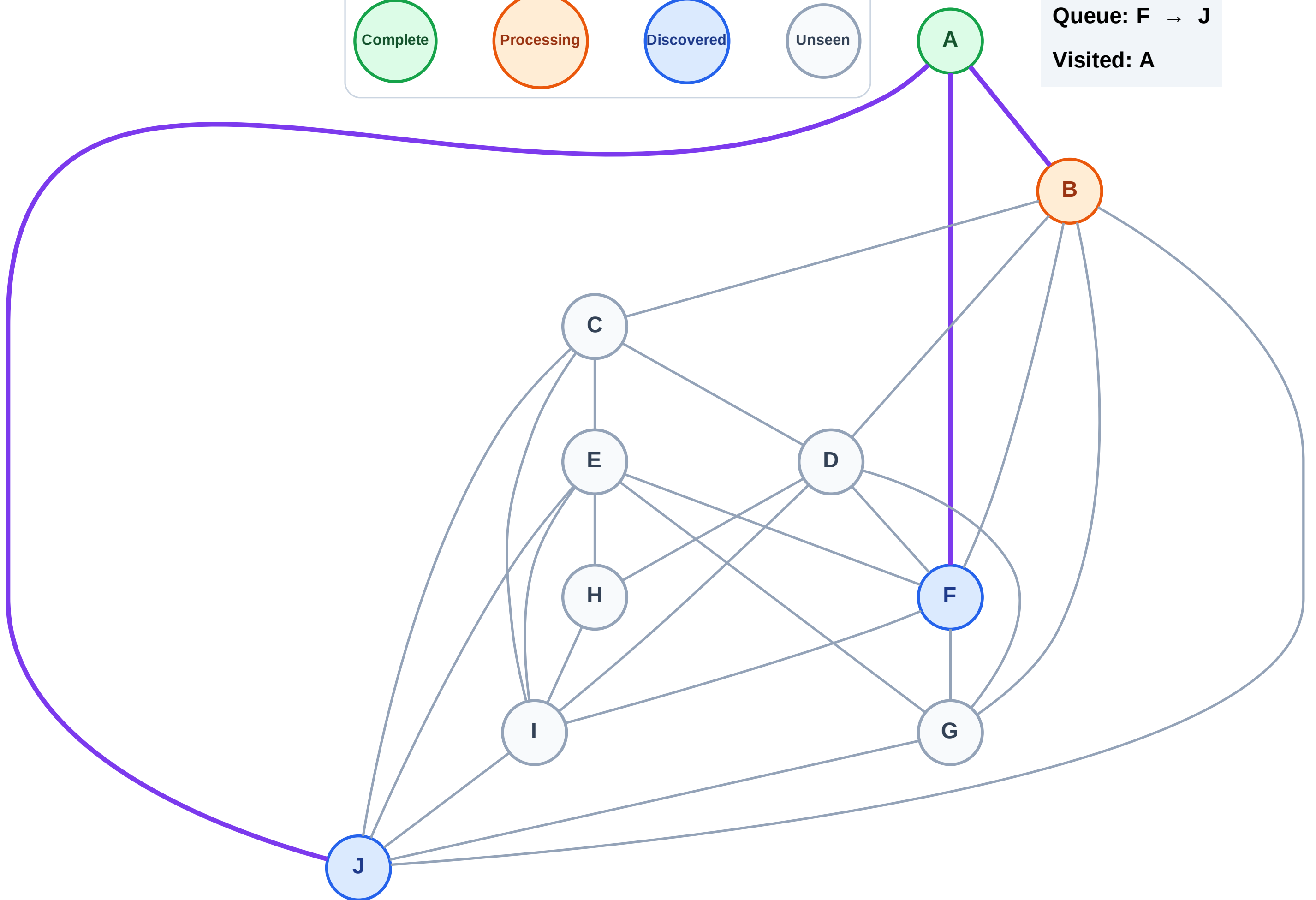
BFS Traversal

Step 4: Process node B

Legend



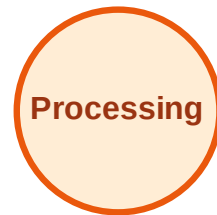
Queue: F → J
Visited: A



BFS Traversal

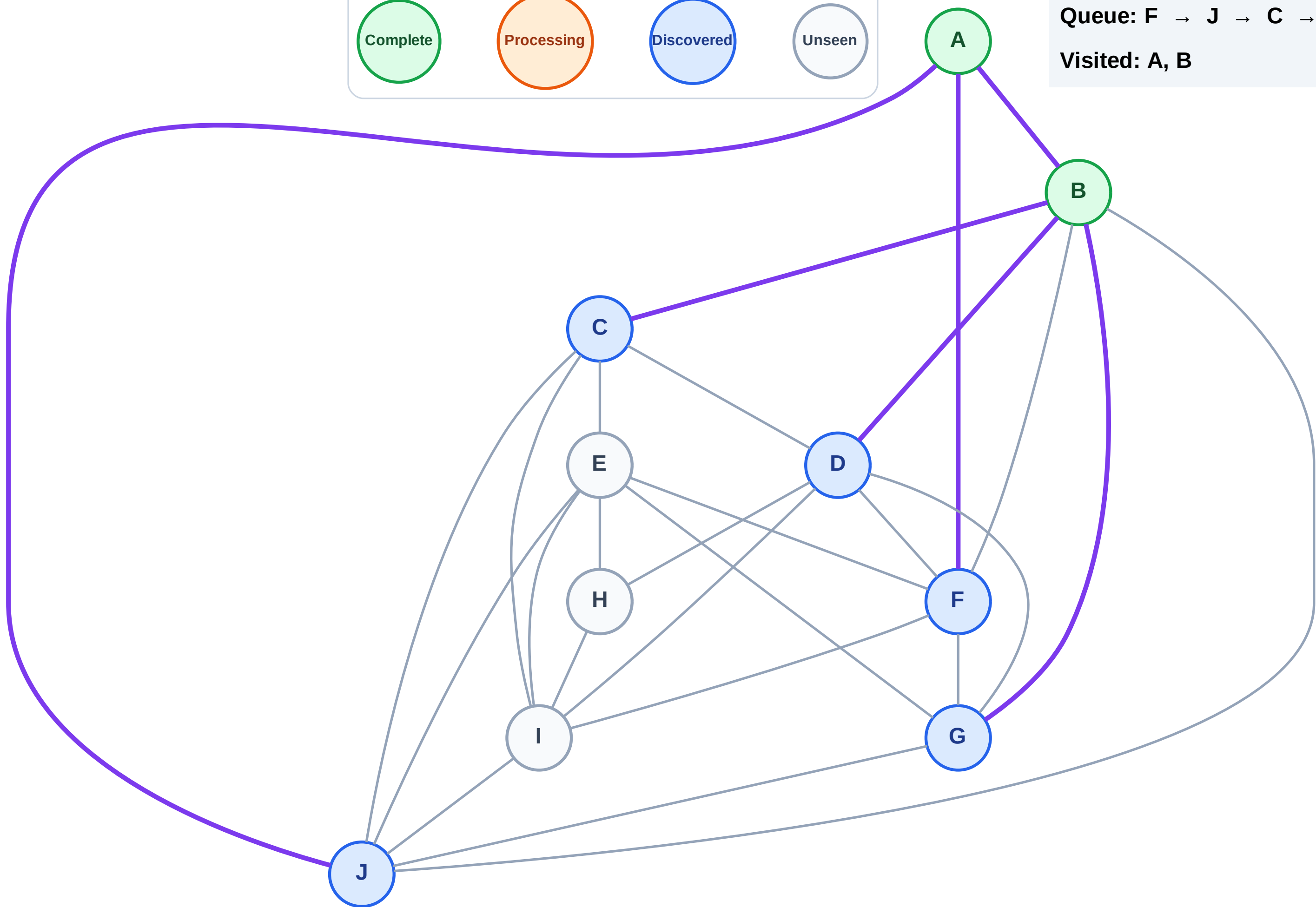
Step 5: Discover C, D, G from B

Legend



Queue: F → J → C → D → G

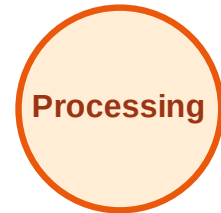
Visited: A, B



BFS Traversal

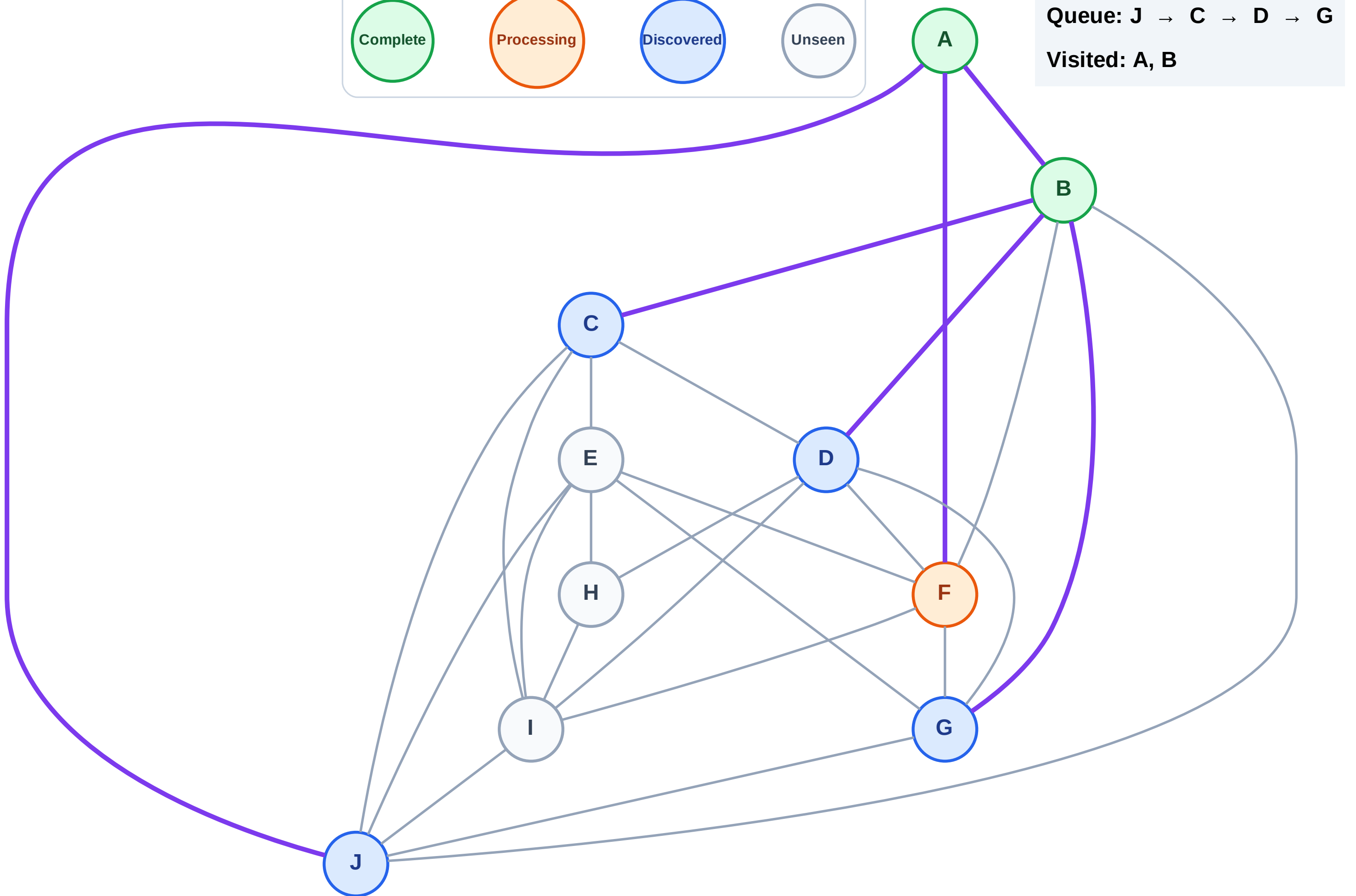
Step 6: Process node F

Legend



Queue: J → C → D → G

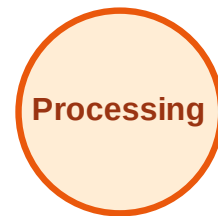
Visited: A, B



BFS Traversal

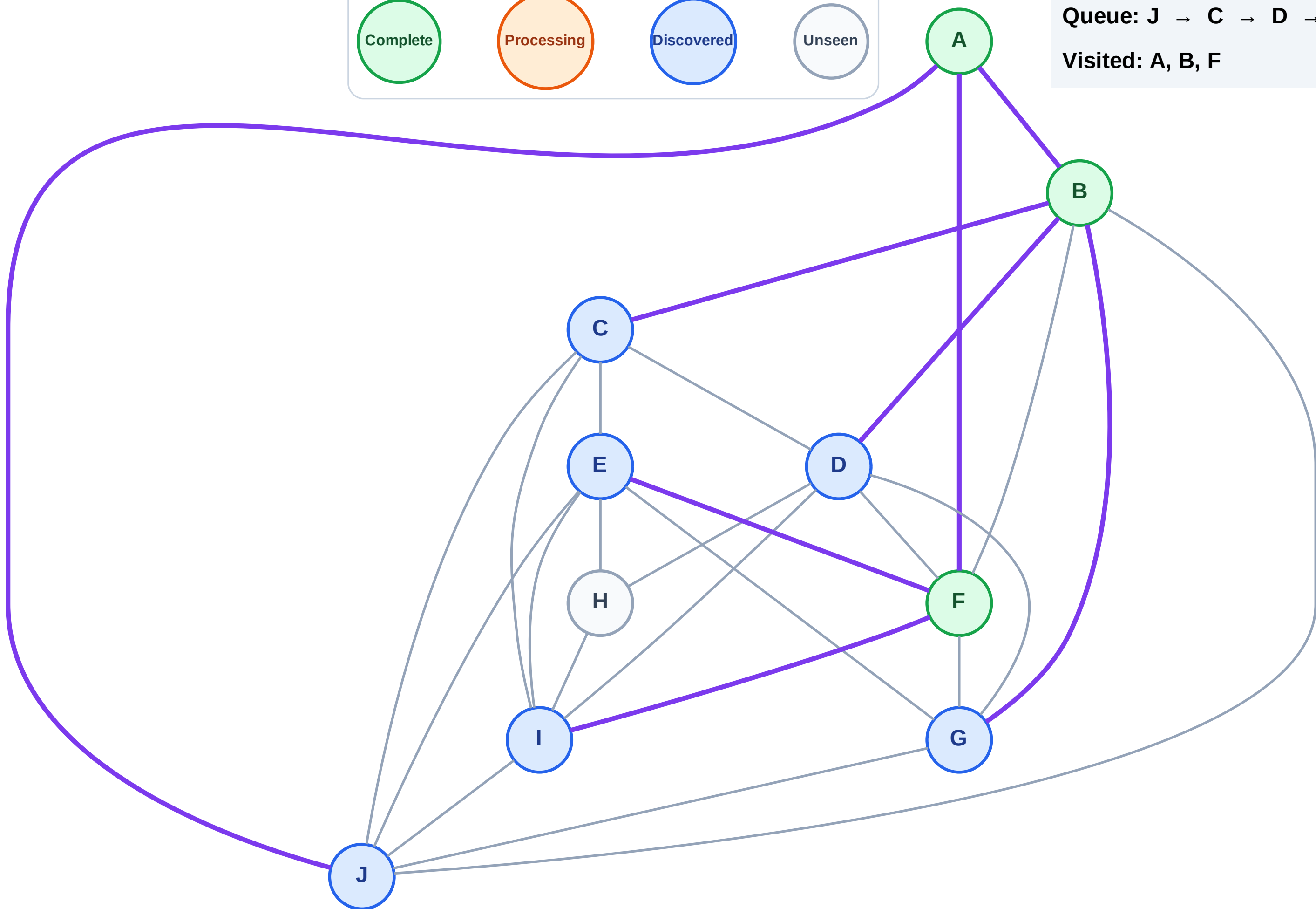
Step 7: Discover E, I from F

Legend



Queue: J → C → D → G → E → I

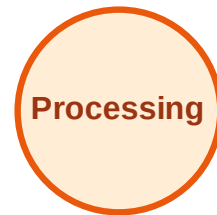
Visited: A, B, F



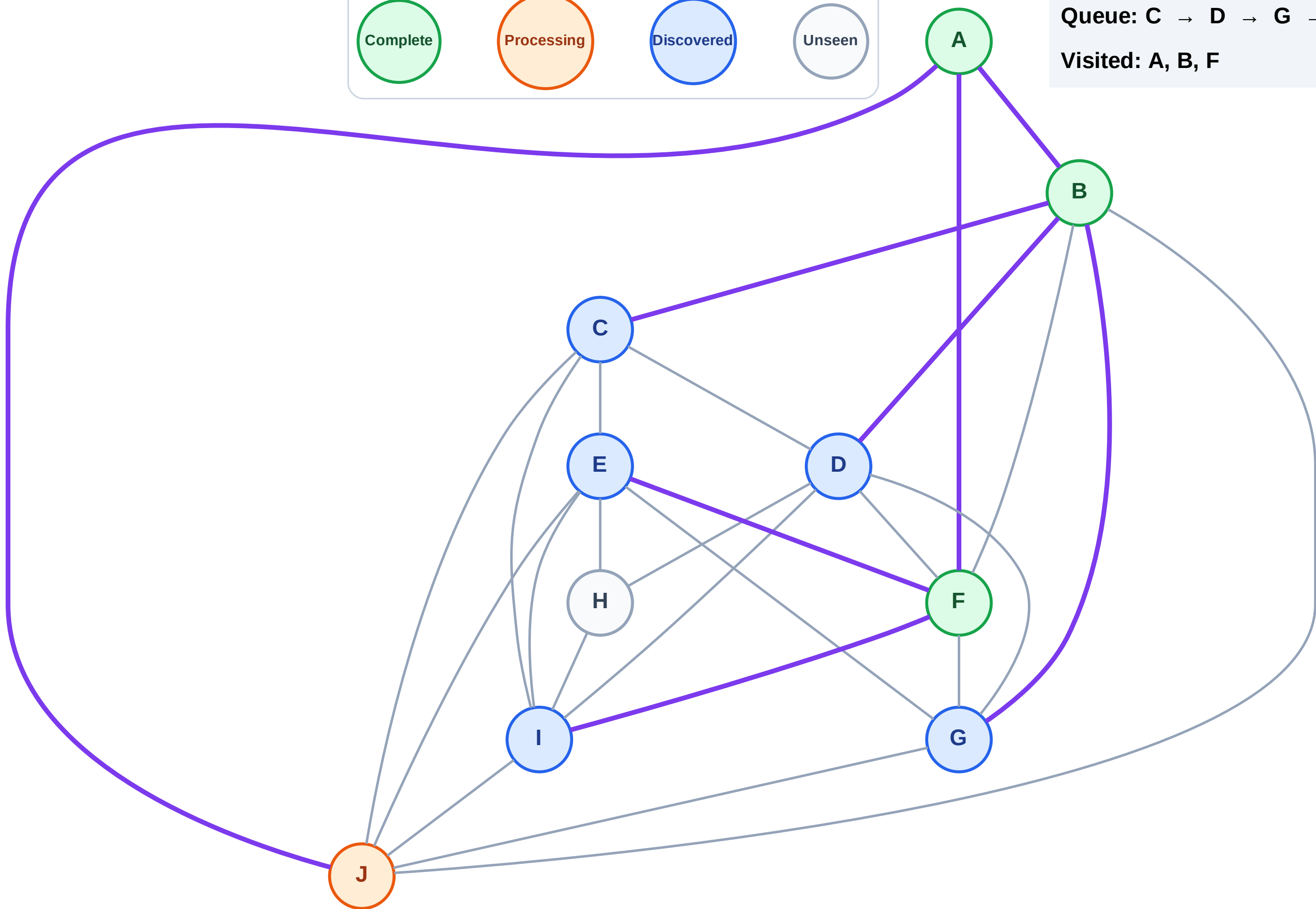
BFS Traversal

Step 8: Process node J

Legend



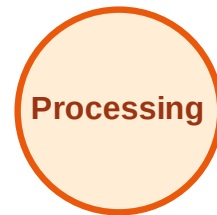
Queue: C → D → G → E → I
Visited: A, B, F



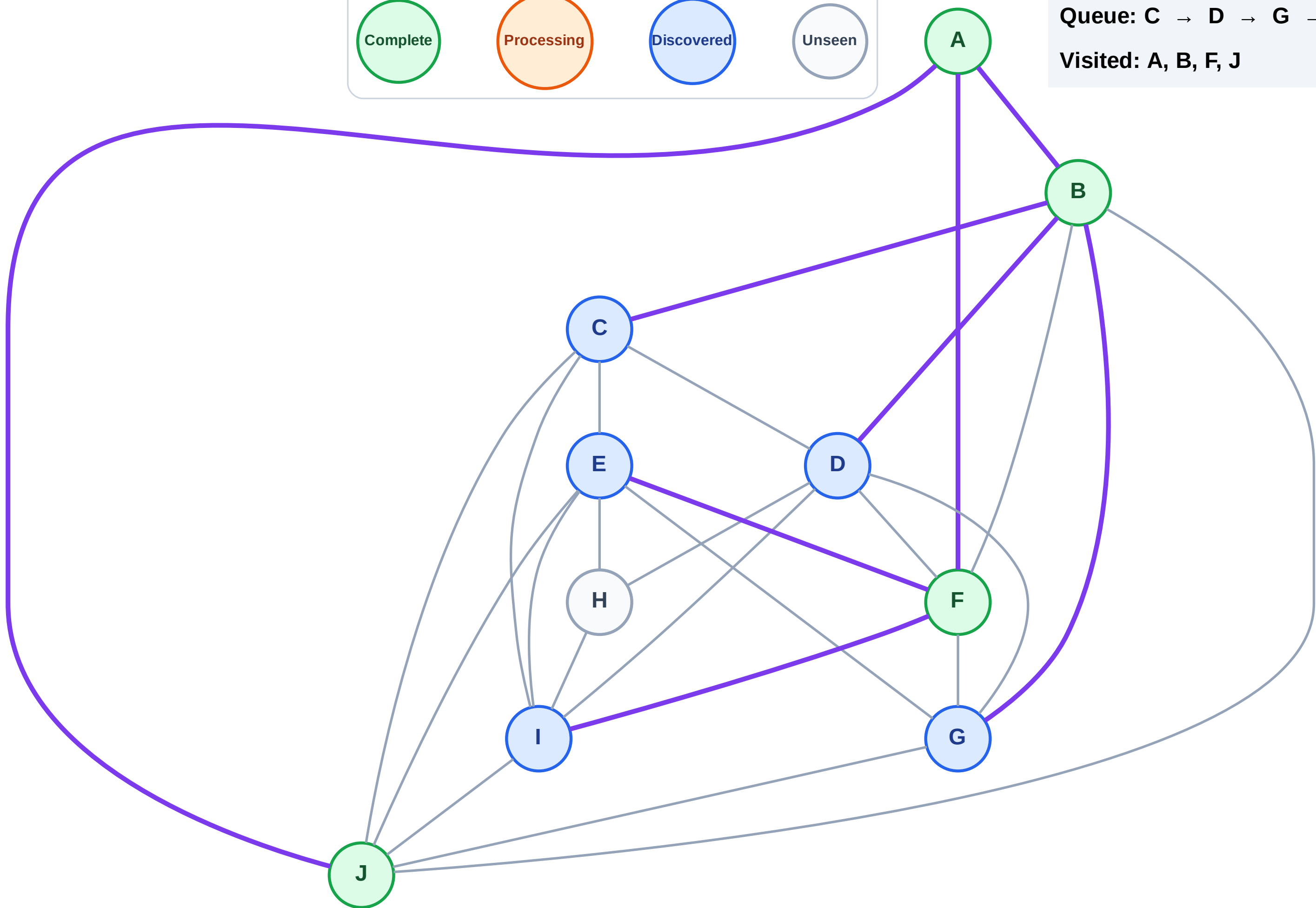
BFS Traversal

Step 9: No new neighbors from J

Legend



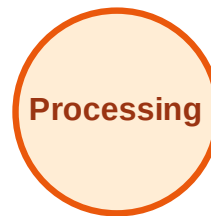
Queue: C → D → G → E → I
Visited: A, B, F, J



BFS Traversal

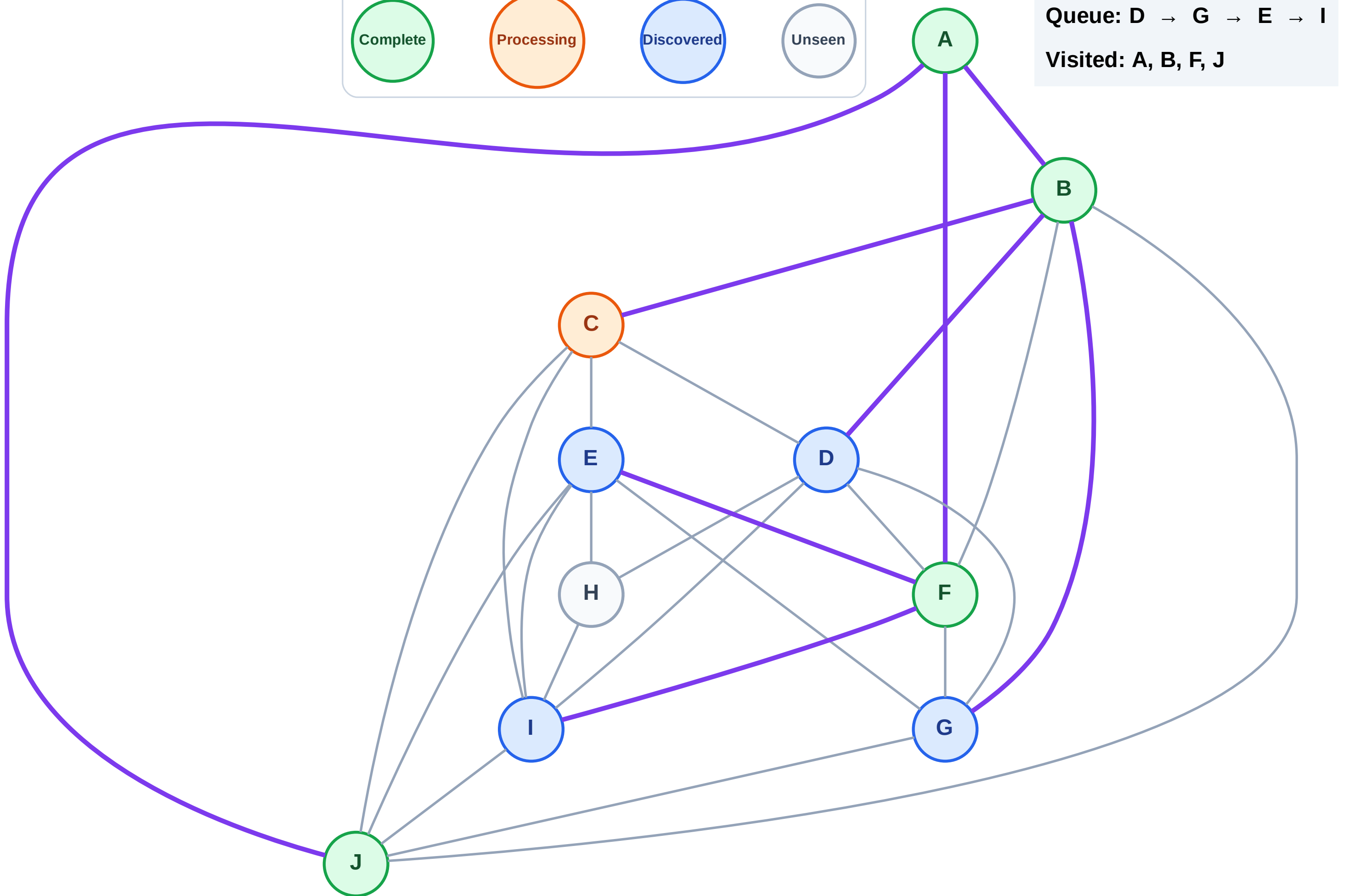
Step 10: Process node C

Legend



Queue: D → G → E → I

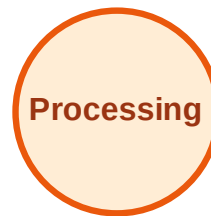
Visited: A, B, F, J



BFS Traversal

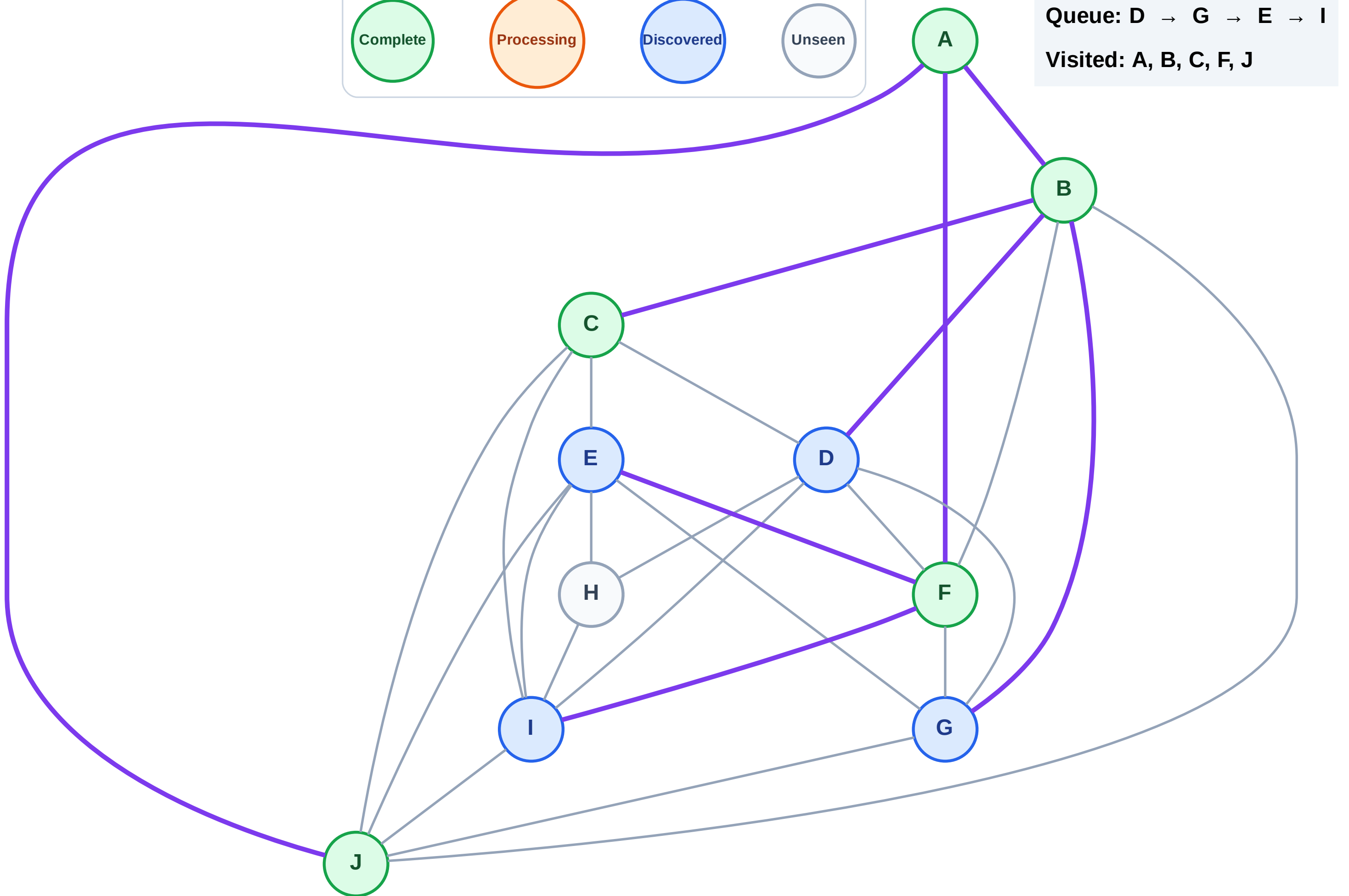
Step 11: No new neighbors from C

Legend



Queue: D → G → E → I

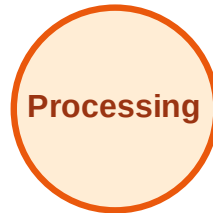
Visited: A, B, C, F, J



BFS Traversal

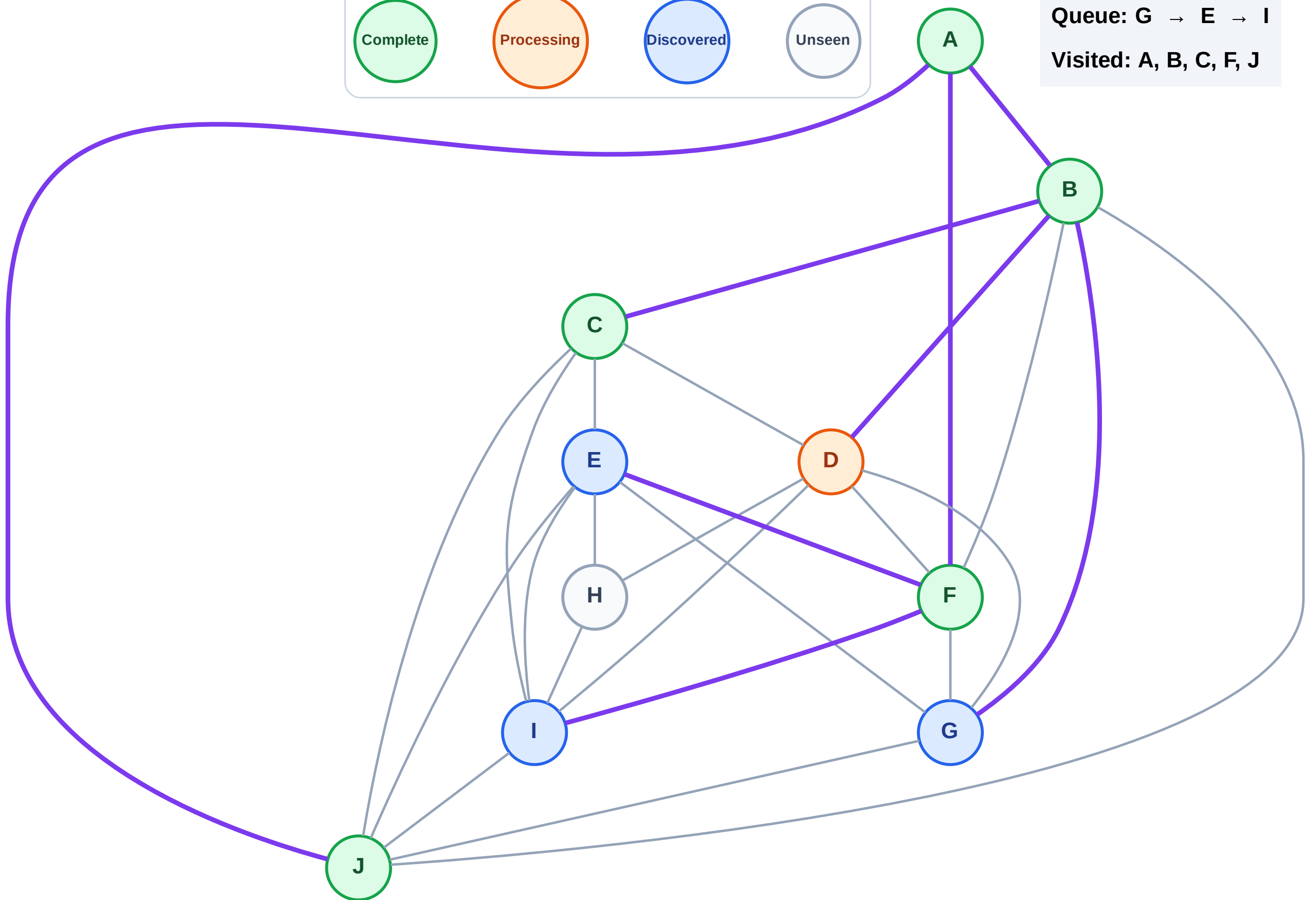
Step 12: Process node D

Legend



Queue: G → E → I

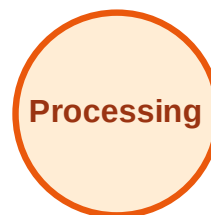
Visited: A, B, C, F, J



BFS Traversal

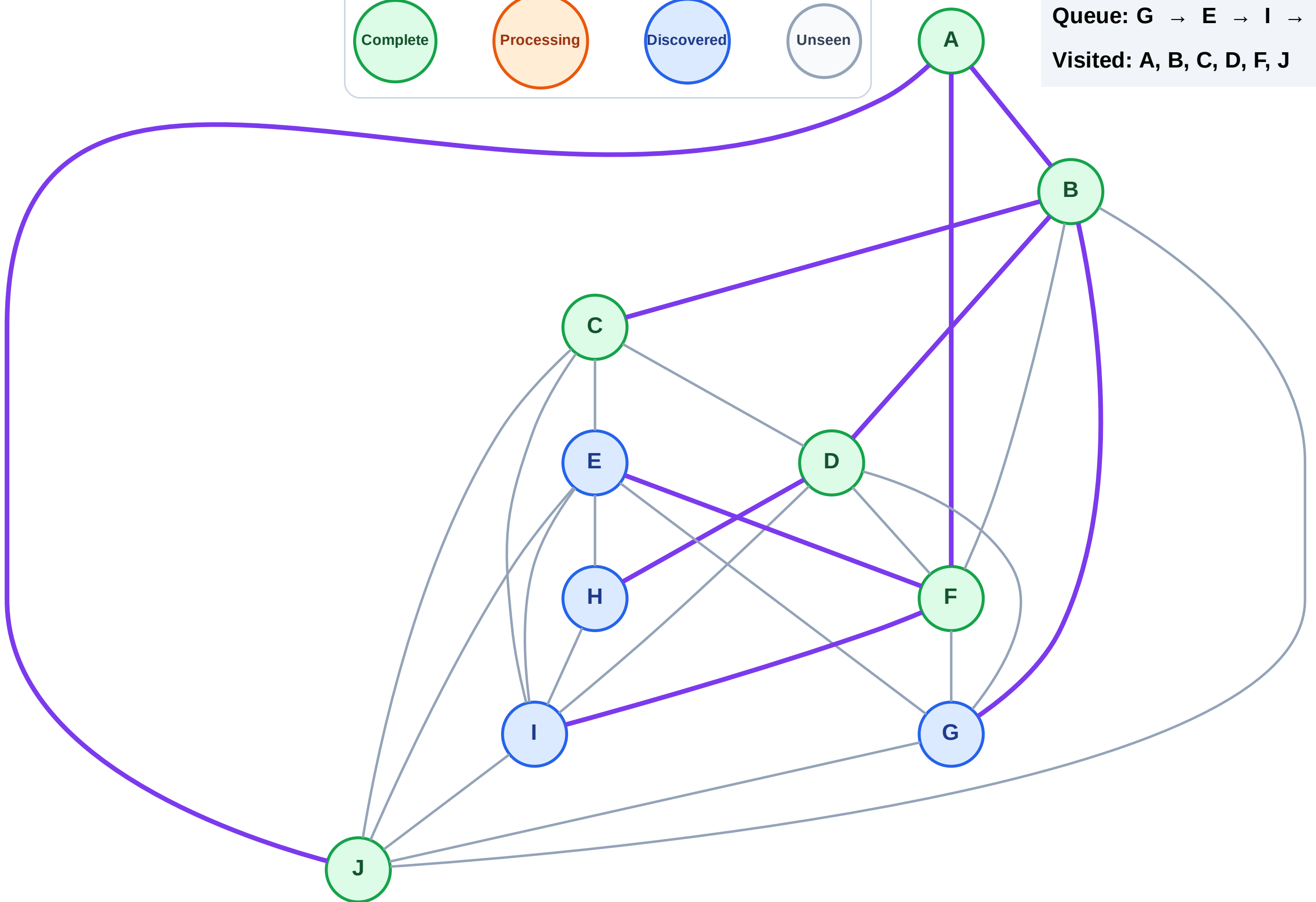
Step 13: Discover H from D

Legend



Queue: G → E → I → H

Visited: A, B, C, D, F, J



Step 14: Process node G

Step 14: Process node G

Legend

Complete

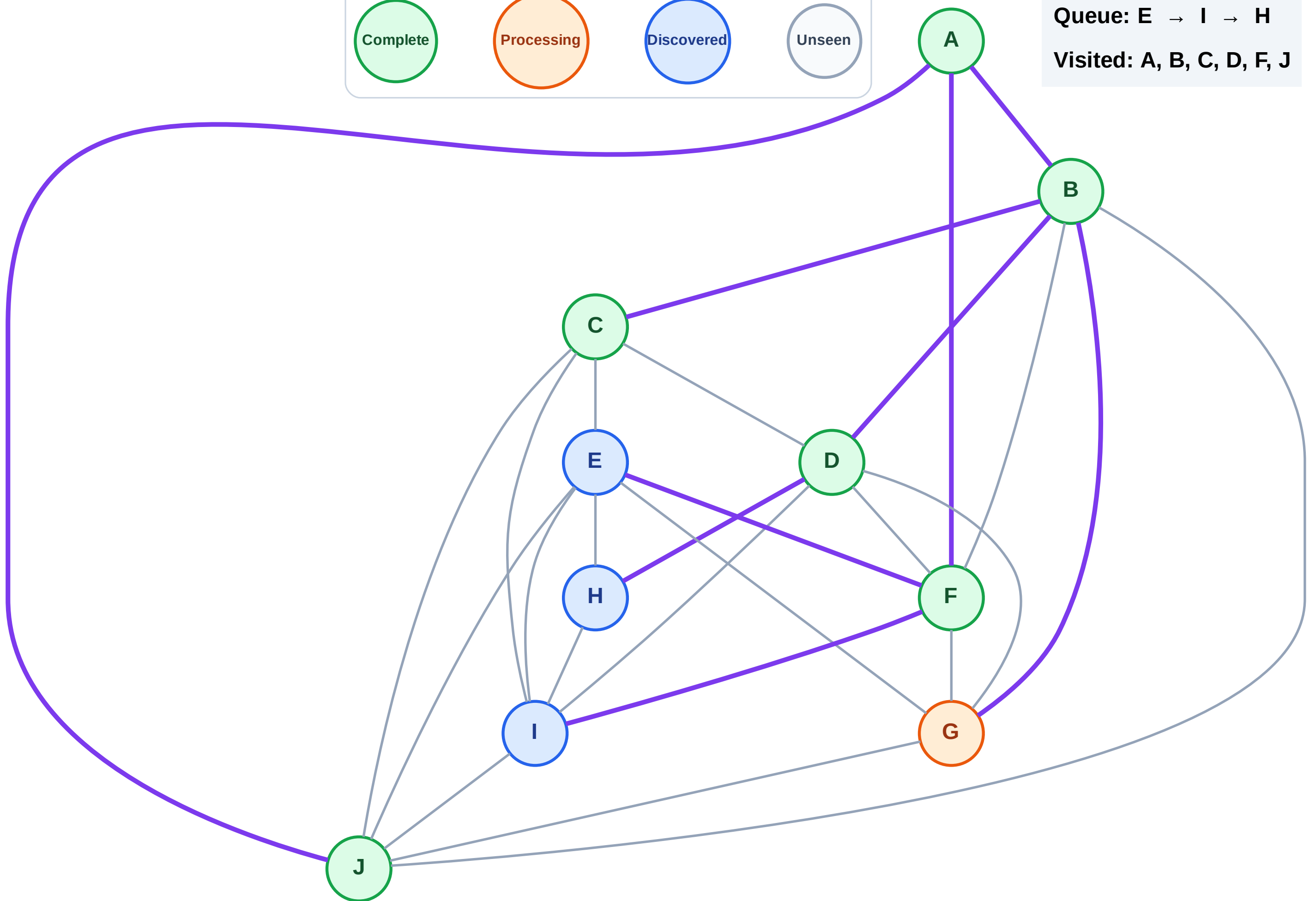
Processing

Discovered

Unseen

Queue: E → I → H

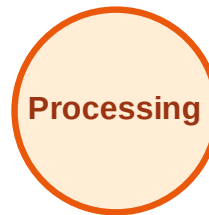
Visited: A, B, C, D, F, J



BFS Traversal

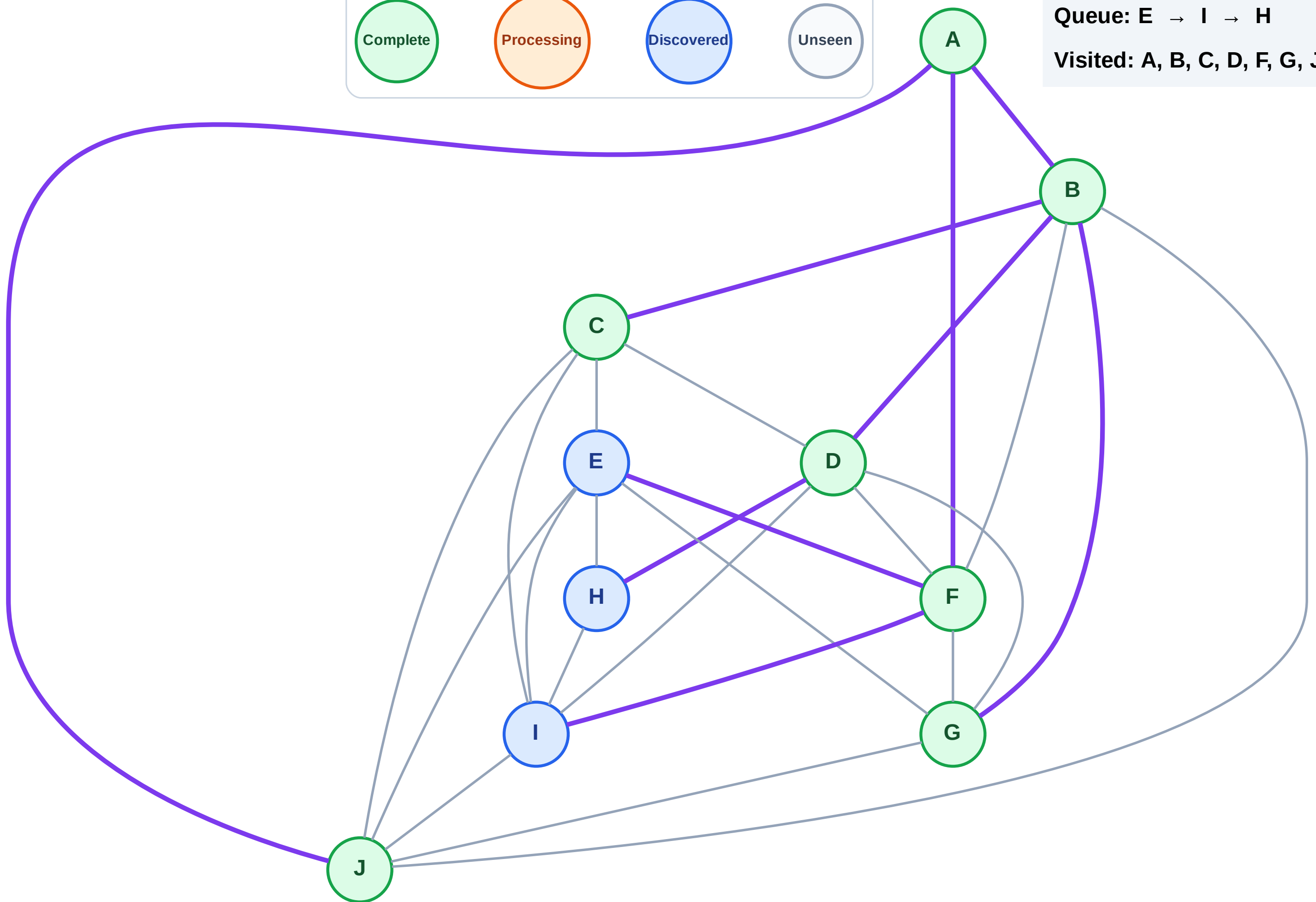
Step 15: No new neighbors from G

Legend



Queue: E → I → H

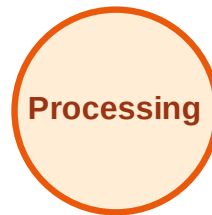
Visited: A, B, C, D, F, G, J



BFS Traversal

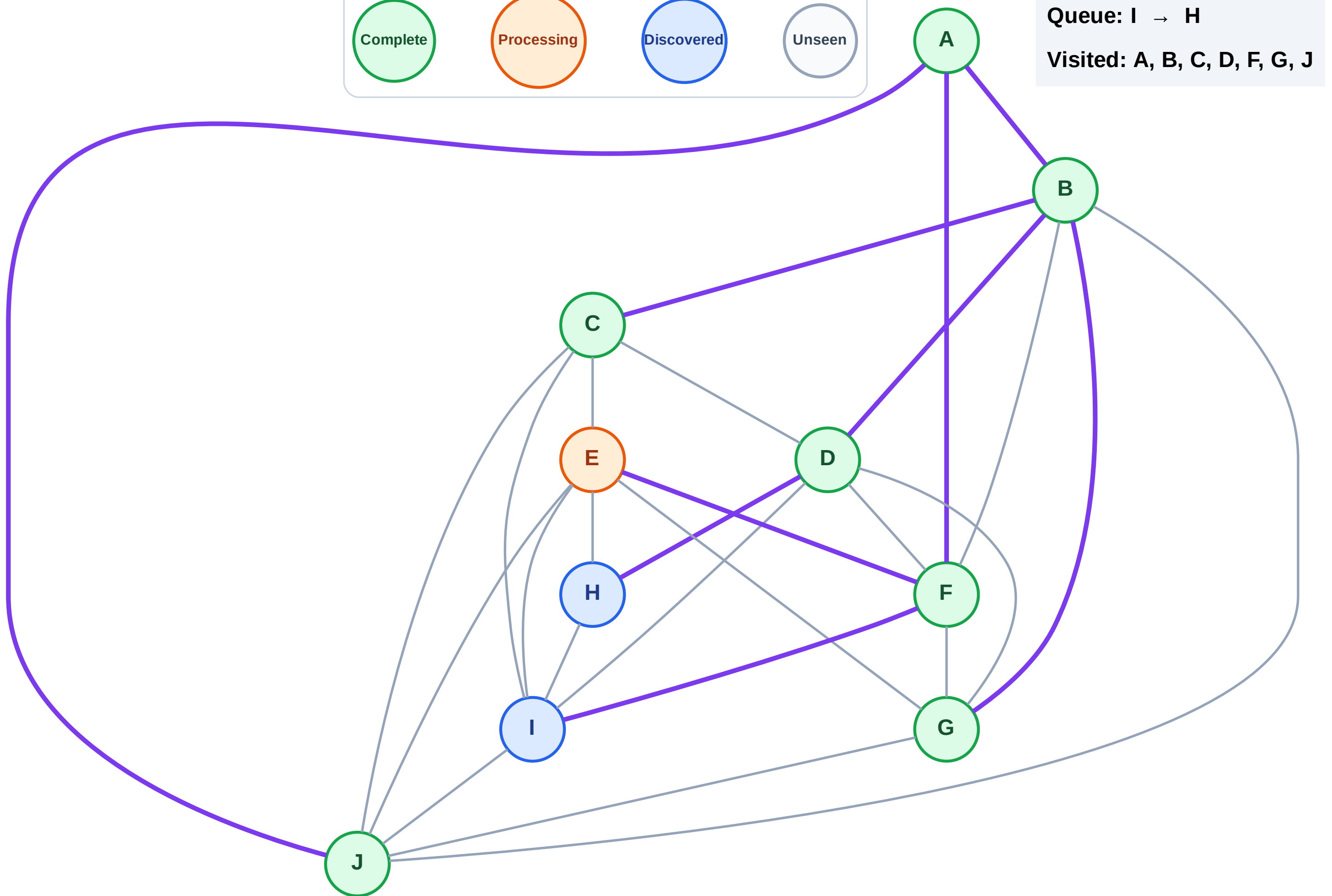
Step 16: Process node E

Legend



Queue: I → H

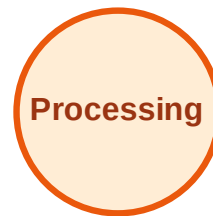
Visited: A, B, C, D, F, G, J



BFS Traversal

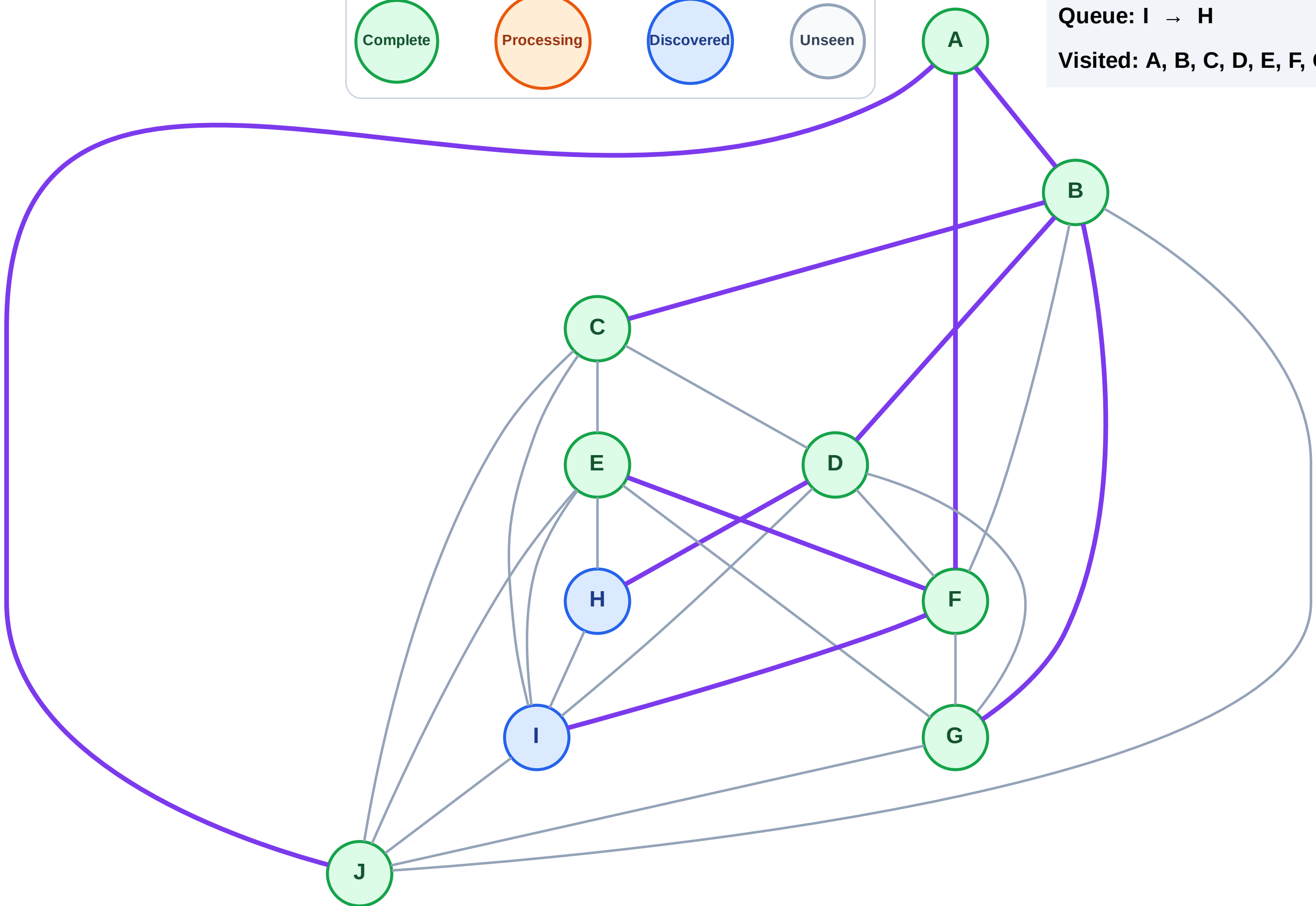
Step 17: No new neighbors from E

Legend



Queue: I → H

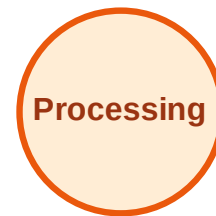
Visited: A, B, C, D, E, F, G, J



BFS Traversal

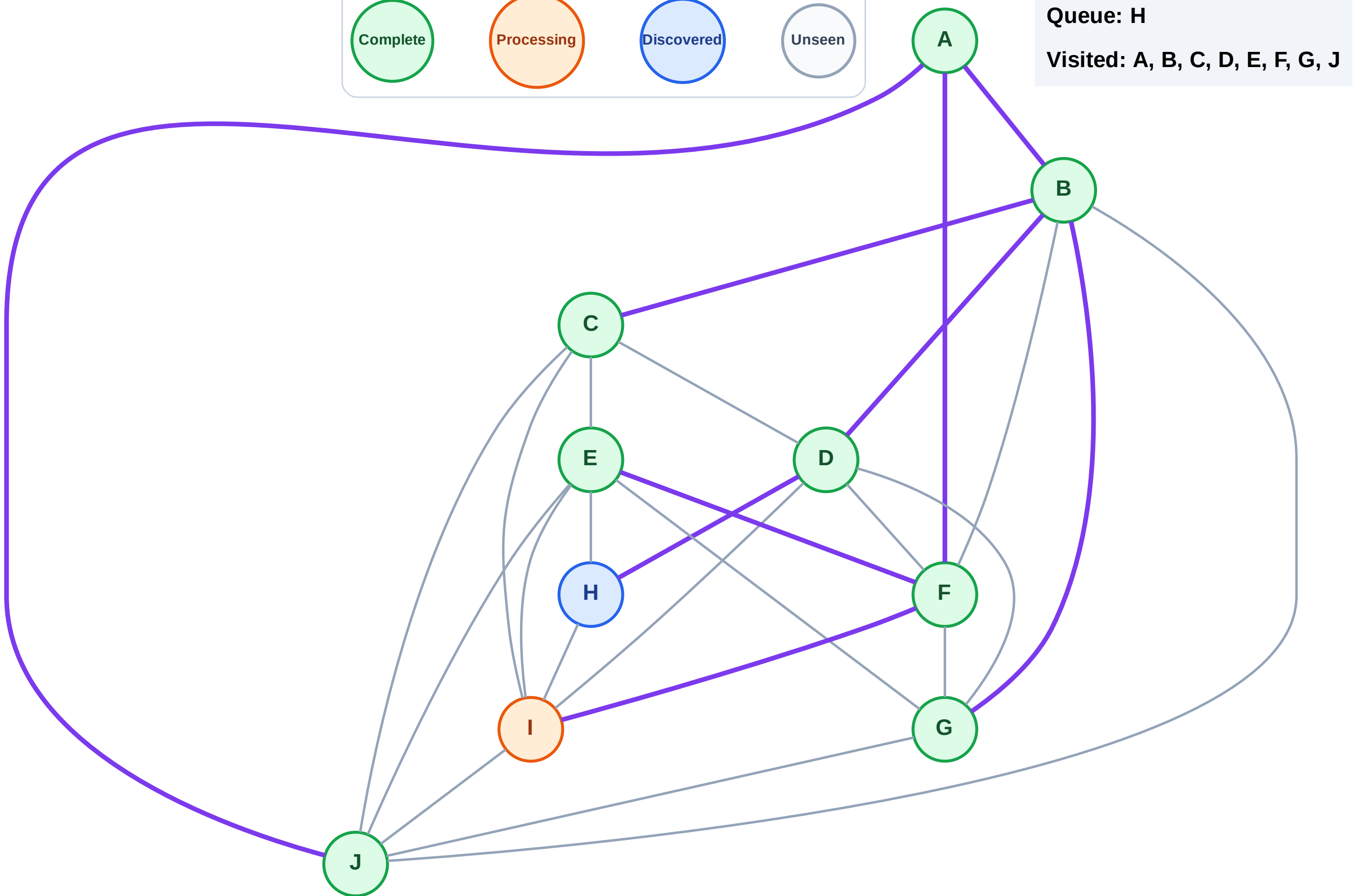
Step 18: Process node I

Legend



Queue: H

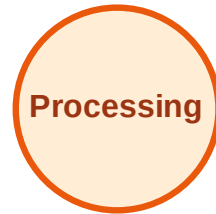
Visited: A, B, C, D, E, F, G, J



BFS Traversal

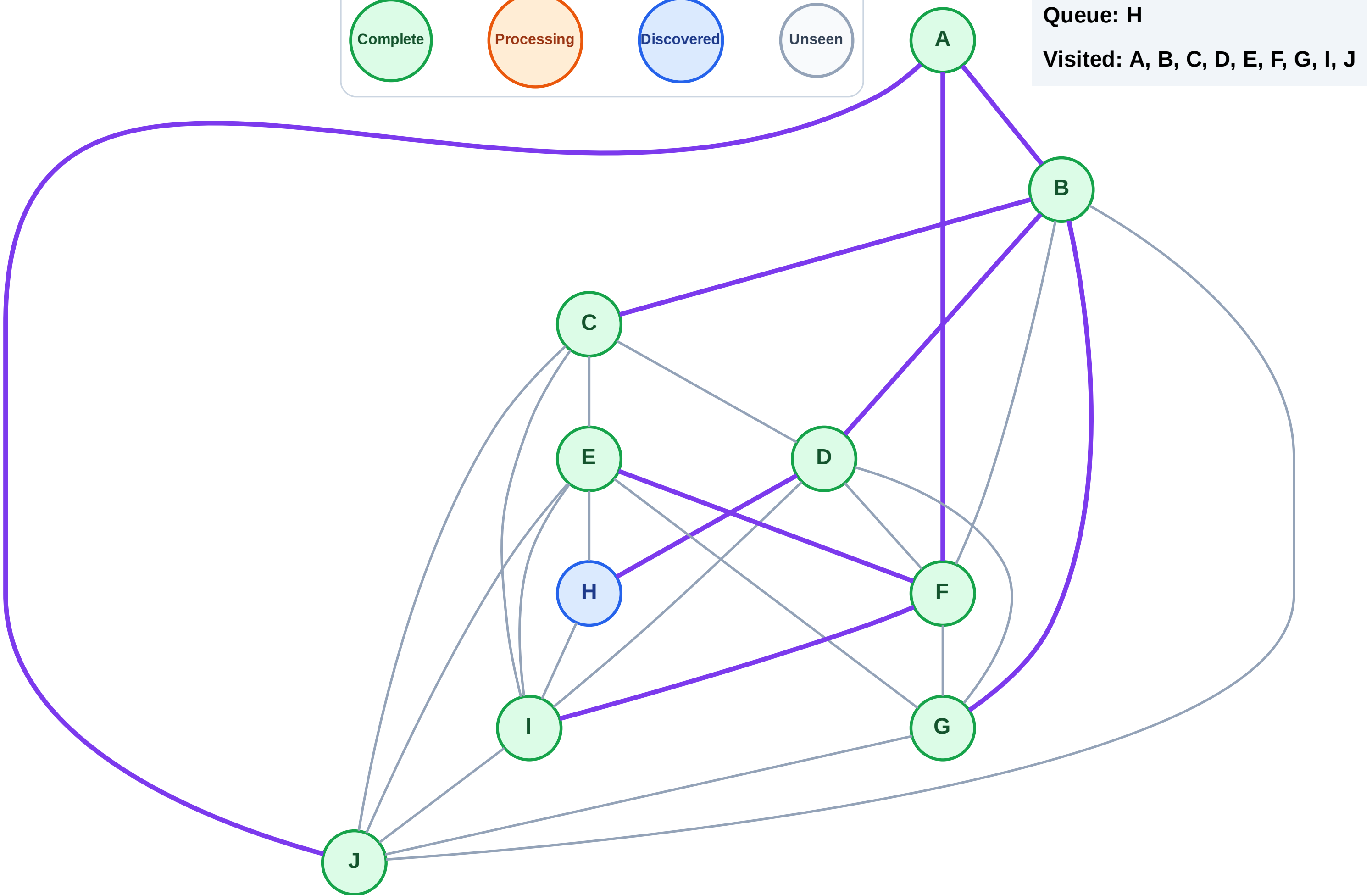
Step 19: No new neighbors from I

Legend



Queue: H

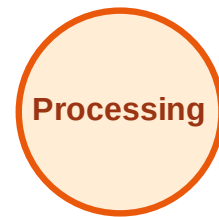
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

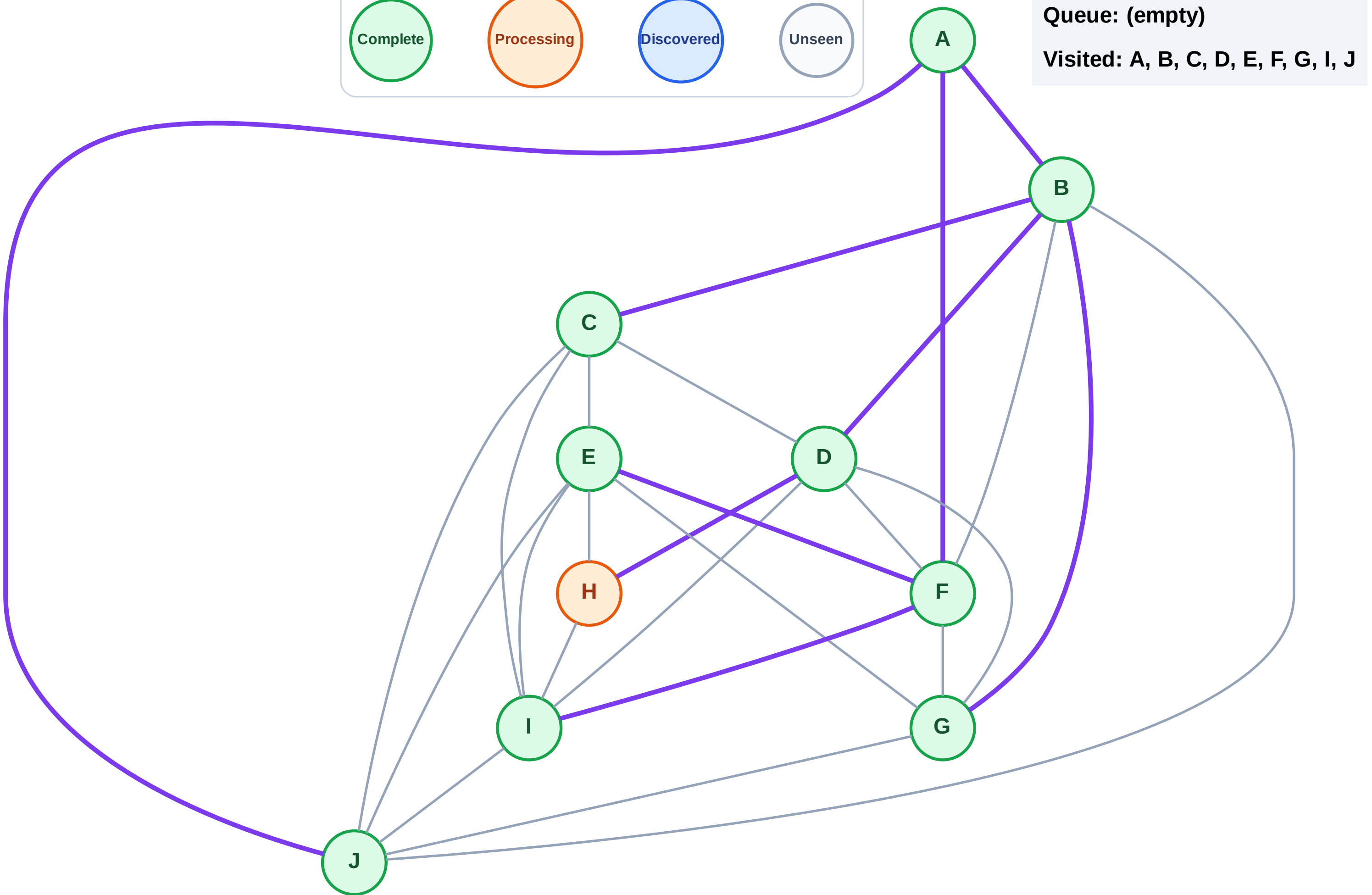
Step 20: Process node H

Legend



Queue: (empty)

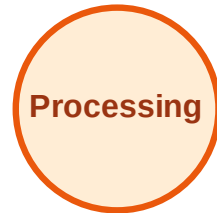
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

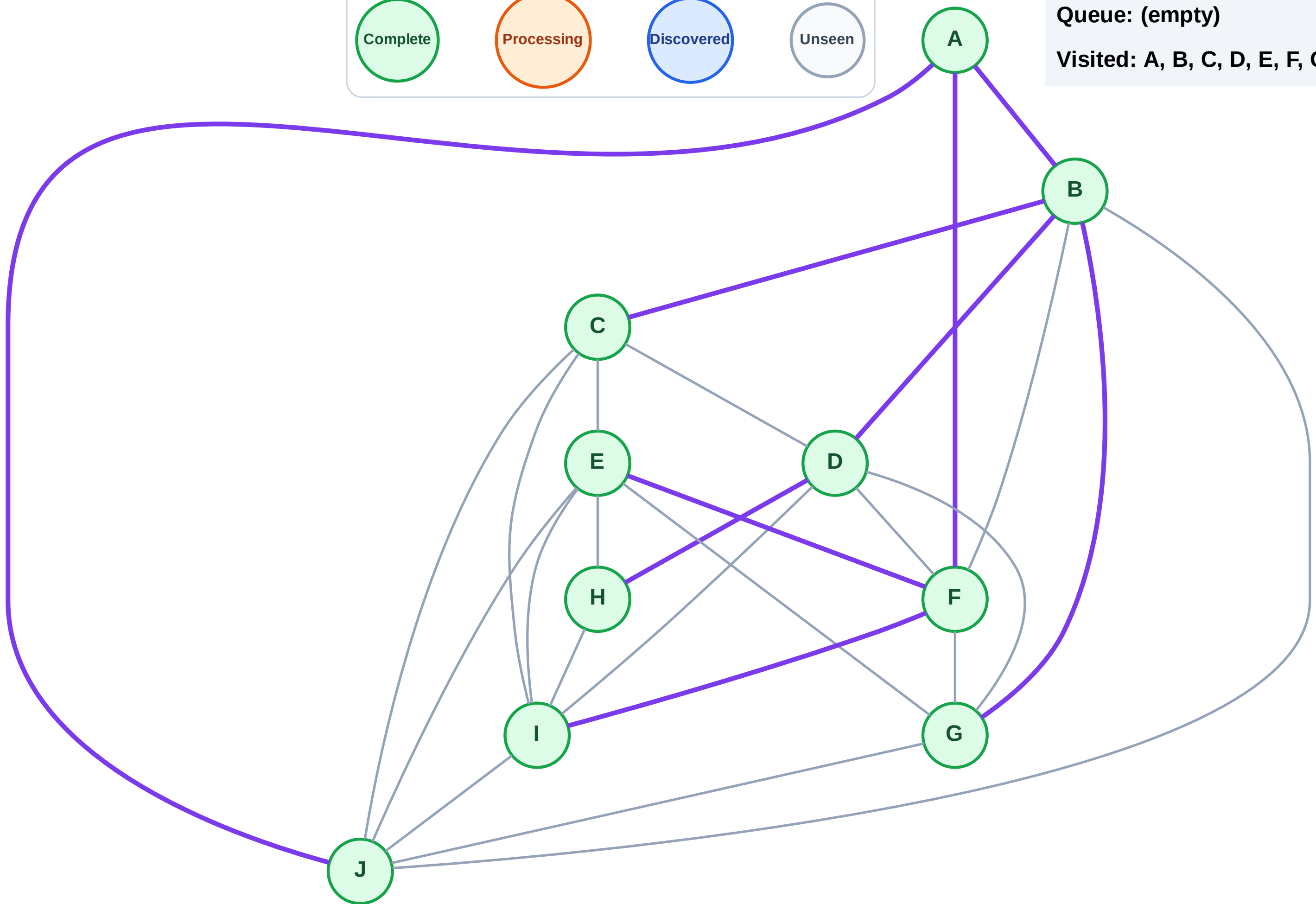
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



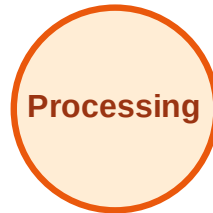
DFS Traversal

Step 1: Start at A

Legend



Complete



Processing



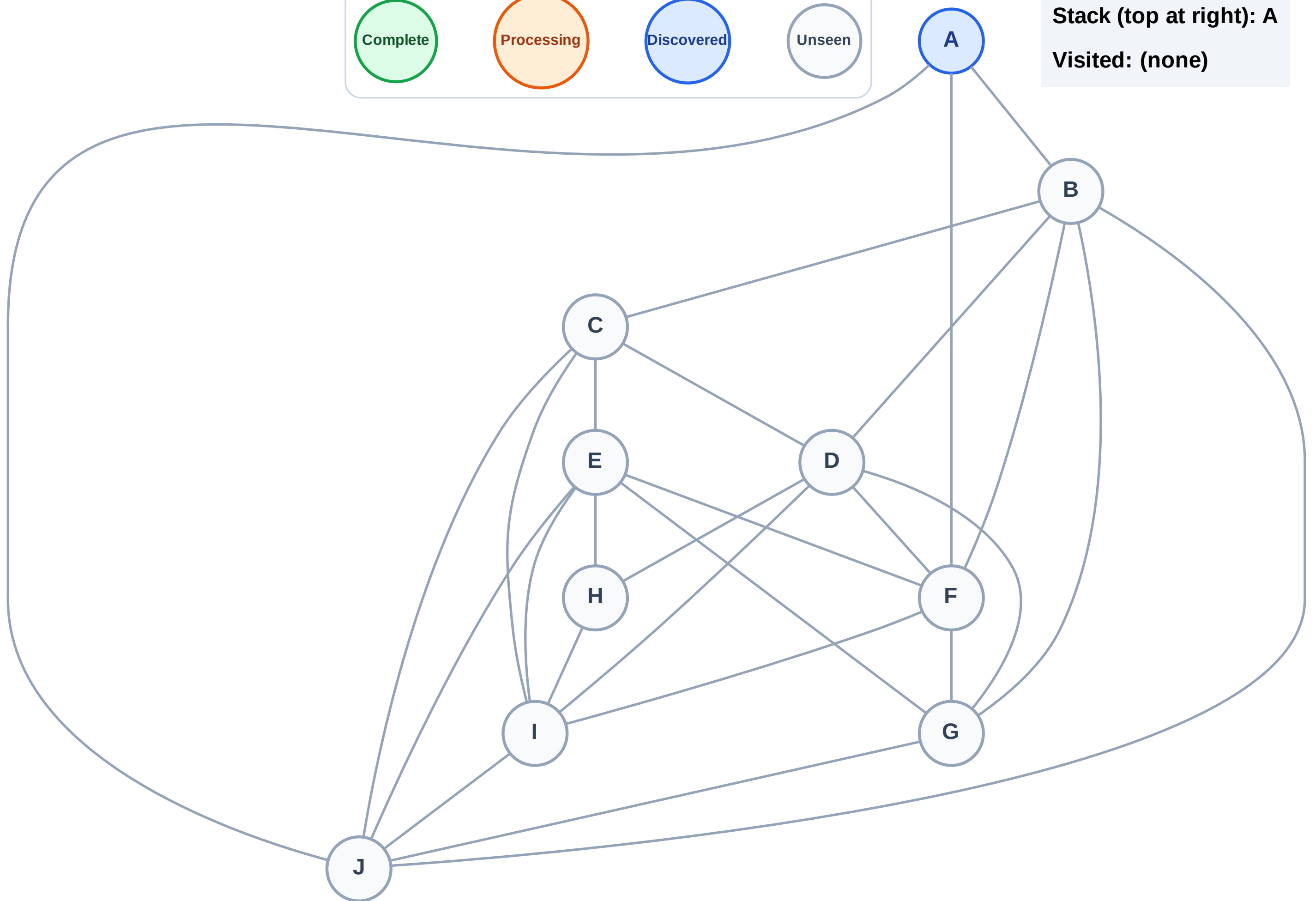
Discovered



Unseen

Stack (top at right): A

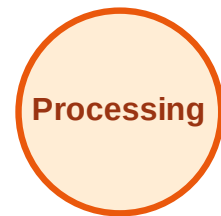
Visited: (none)



DFS Traversal

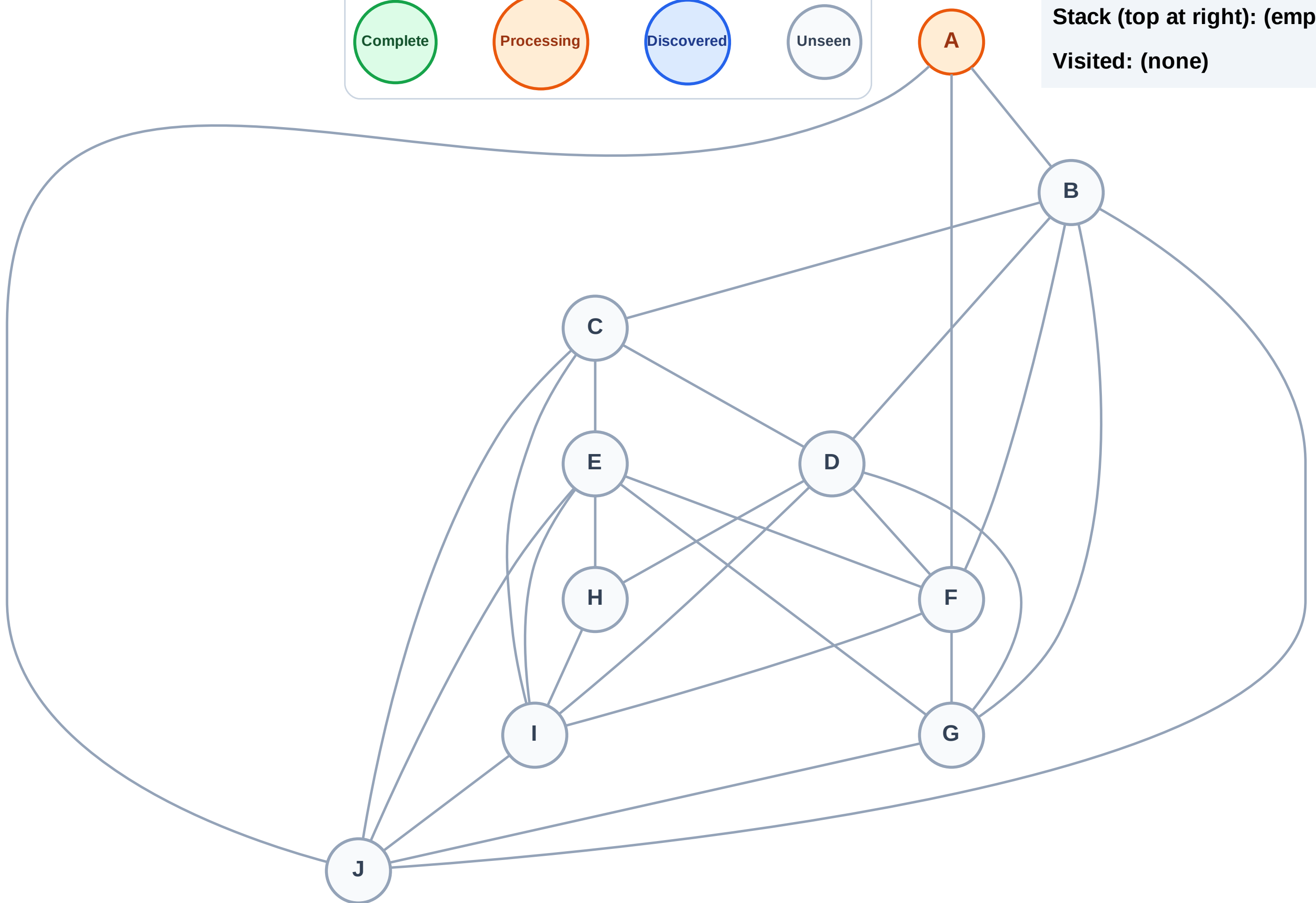
Step 2: Process node A

Legend



Stack (top at right): (empty)

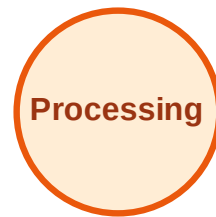
Visited: (none)



DFS Traversal

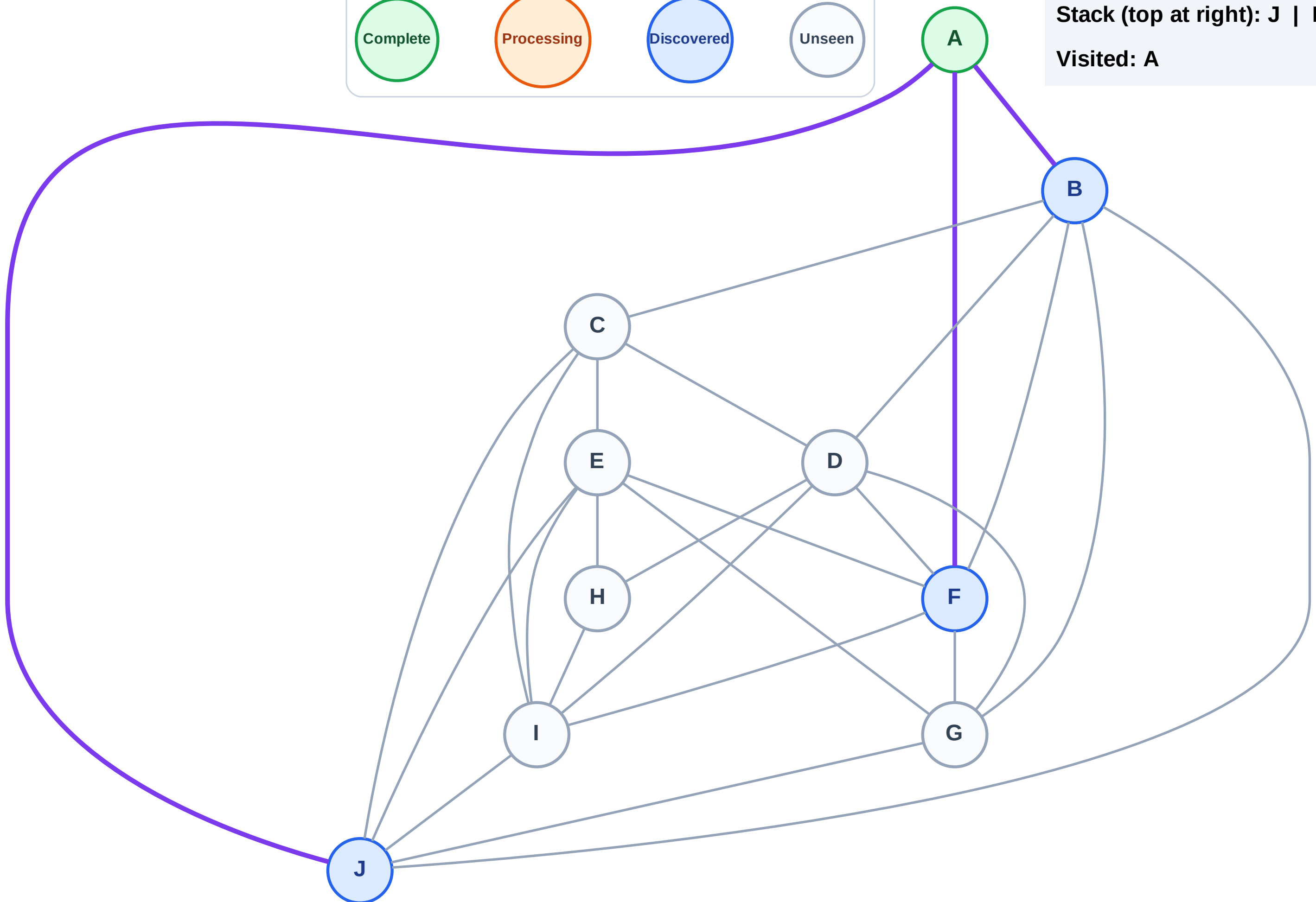
Step 3: Discover J, F, B from A

Legend



Stack (top at right): J | F | B

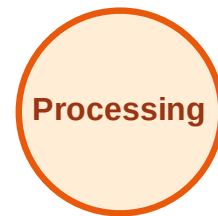
Visited: A



DFS Traversal

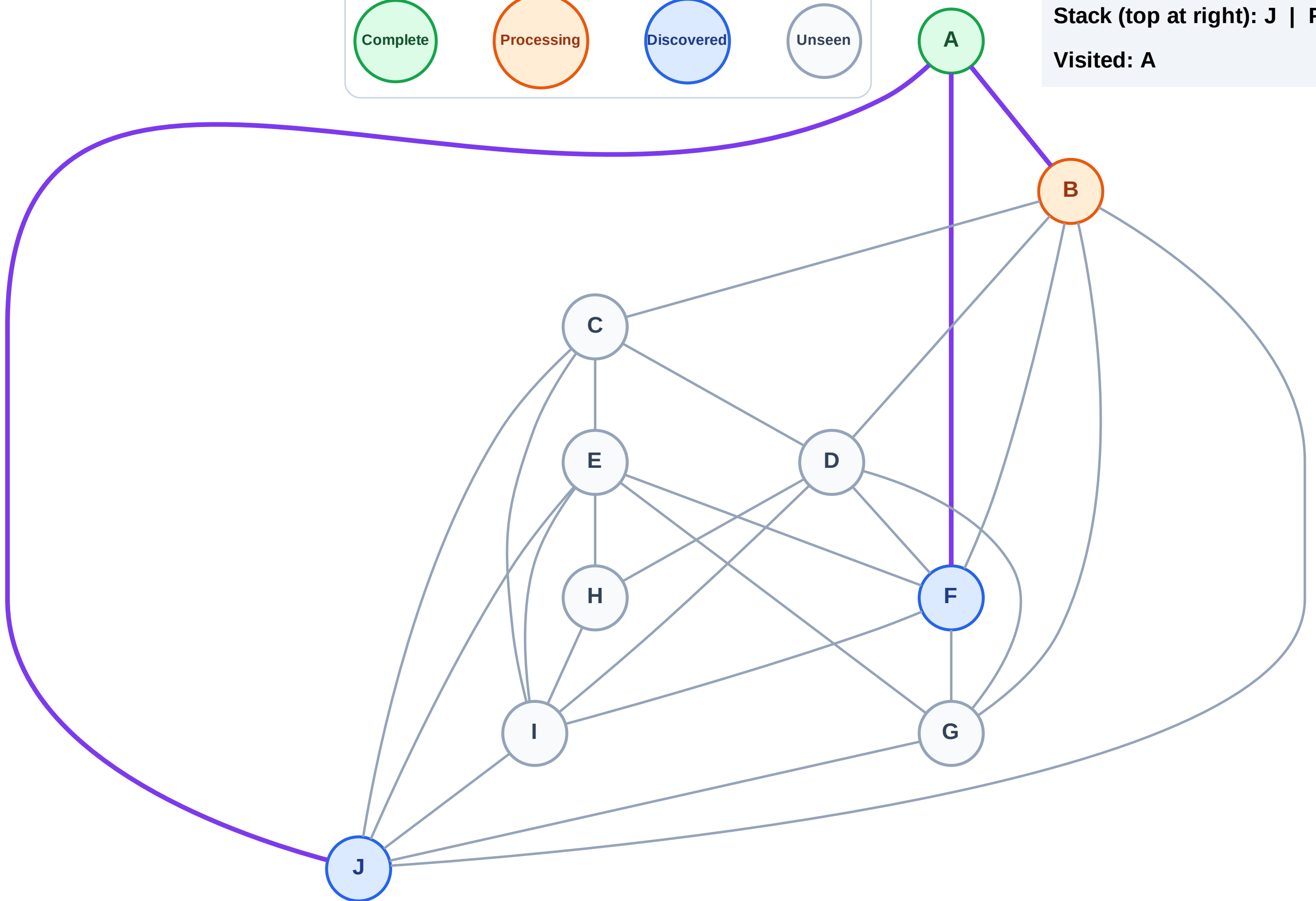
Step 4: Process node B

Legend



Stack (top at right): J | F

Visited: A



DFS Traversal

Step 5: Discover G, D, C from B

Legend

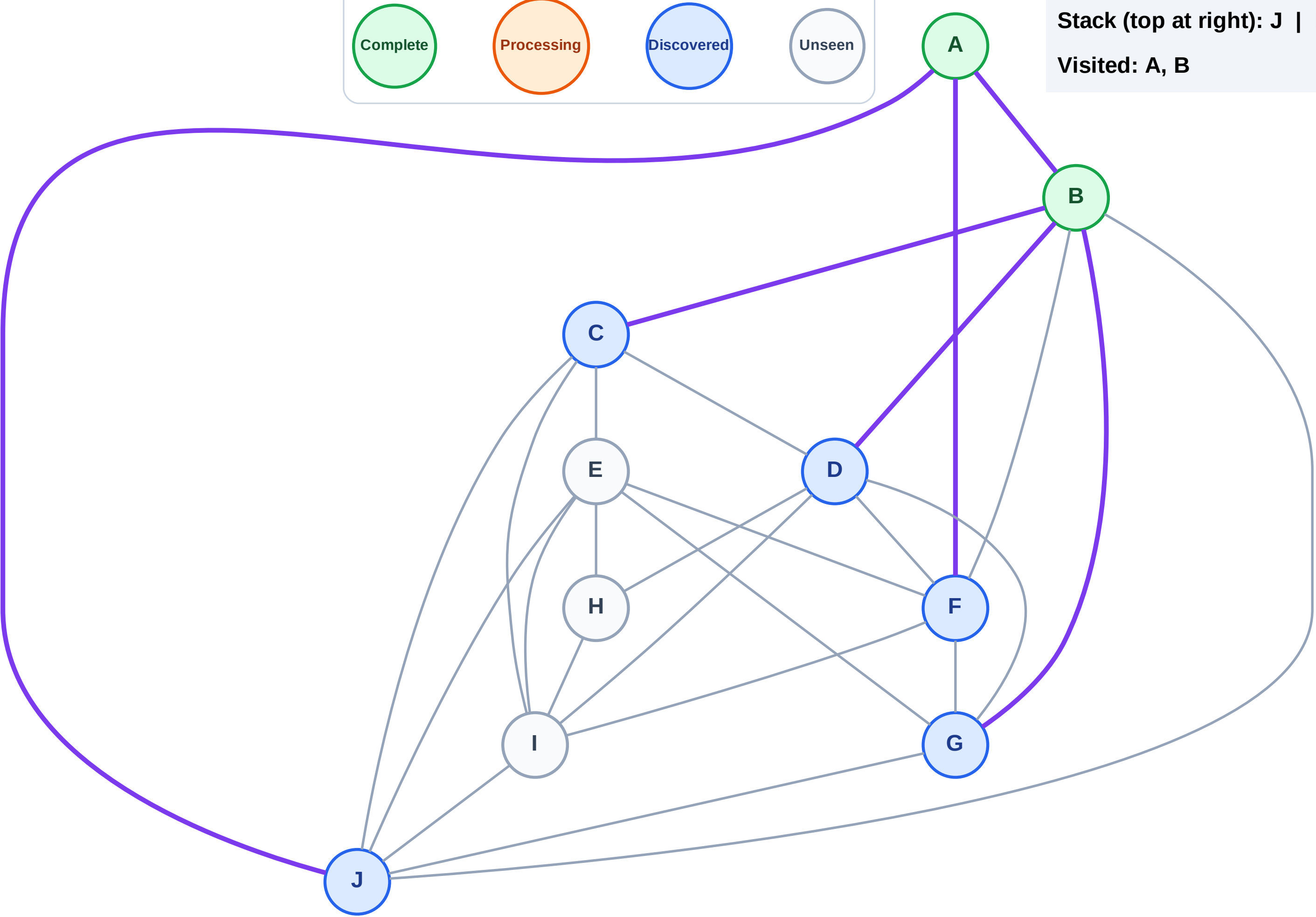
Complete

Processing

Discovered

Unseen

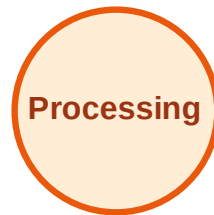
Stack (top at right): J | F | G | D | C
Visited: A, B



DFS Traversal

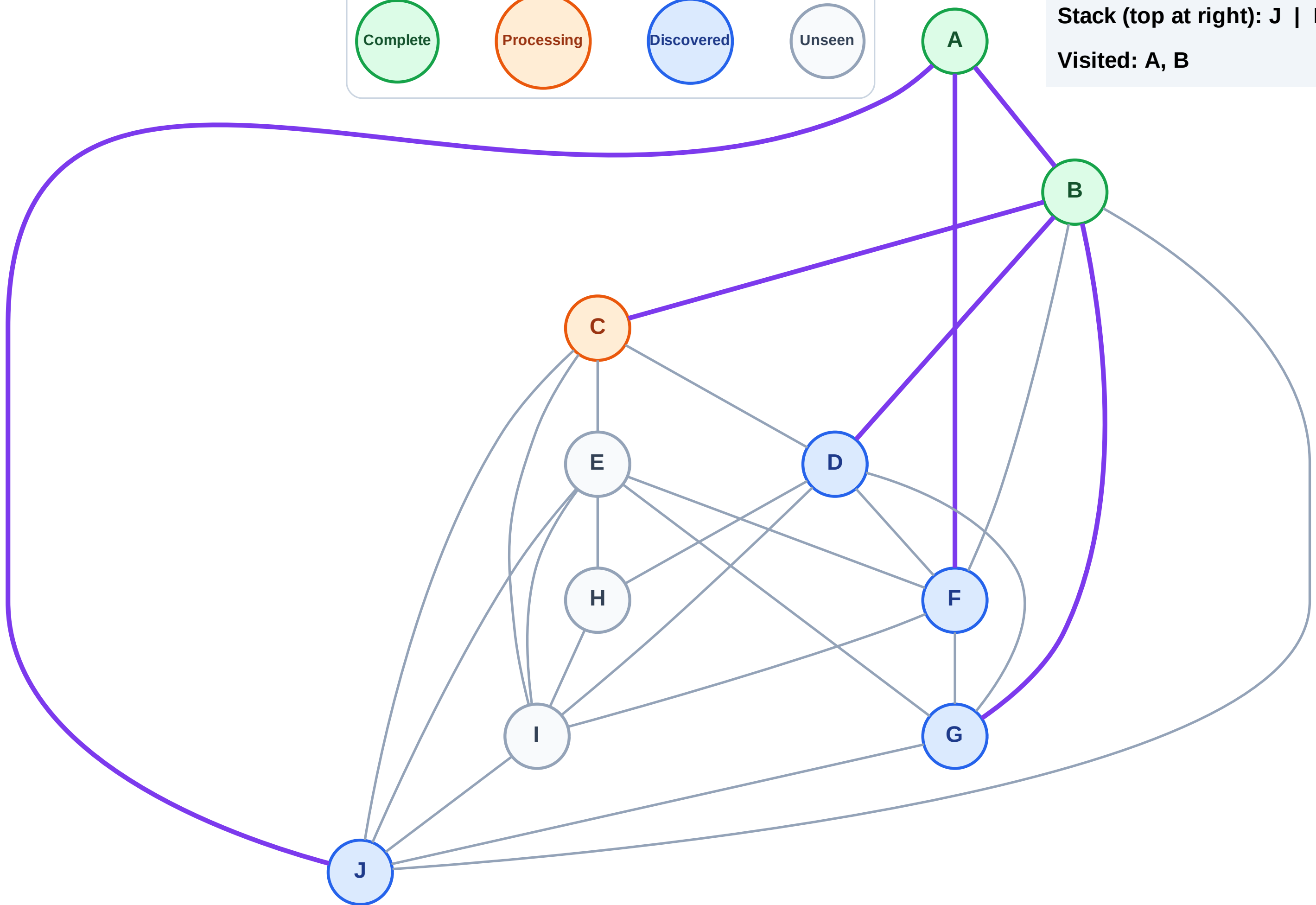
Step 6: Process node C

Legend



Stack (top at right): J | F | G | D

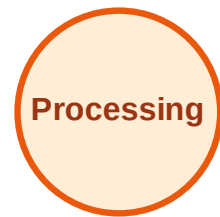
Visited: A, B



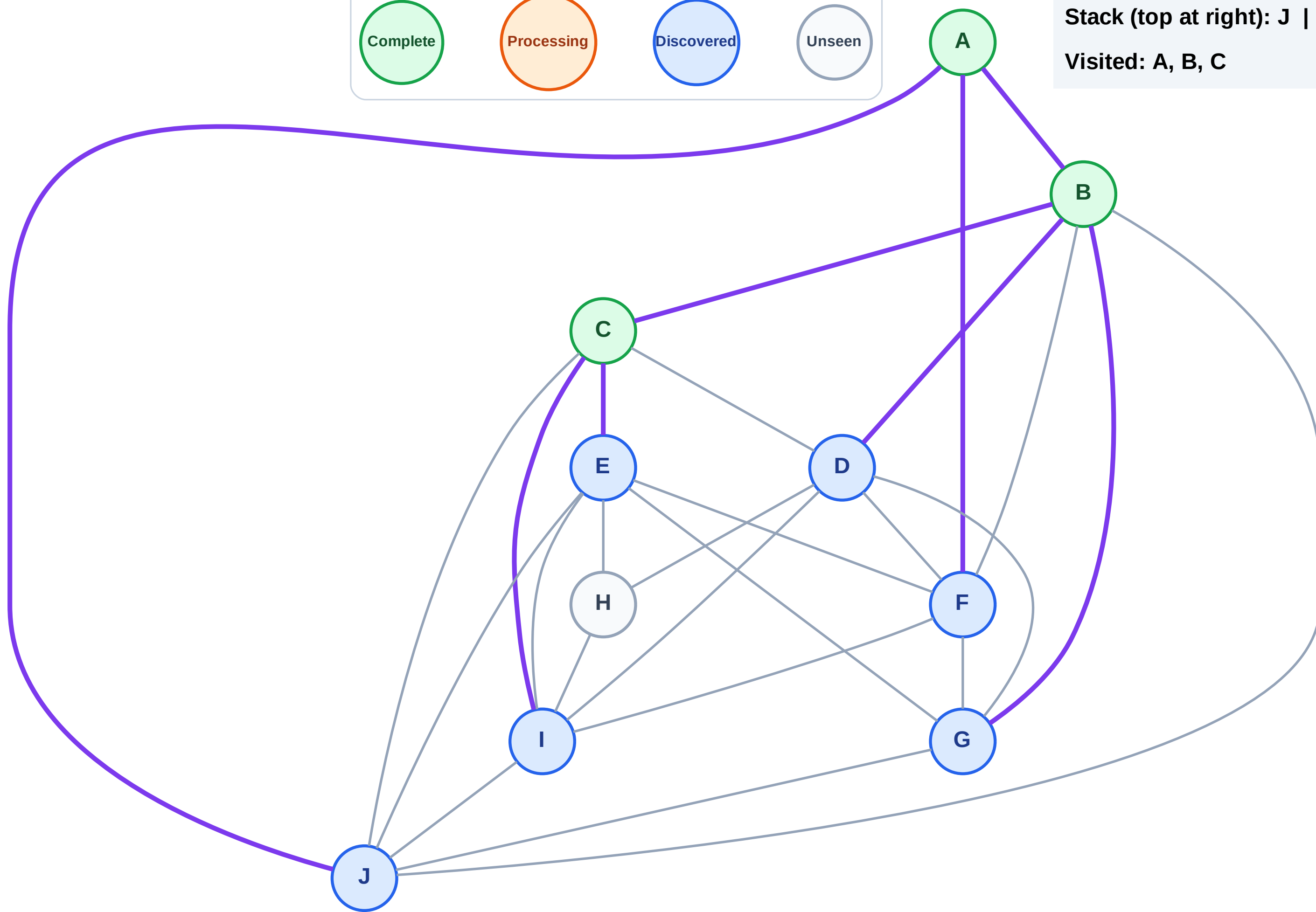
DFS Traversal

Step 7: Discover I, E from C

Legend



Stack (top at right): J | F | G | D | I | E
Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

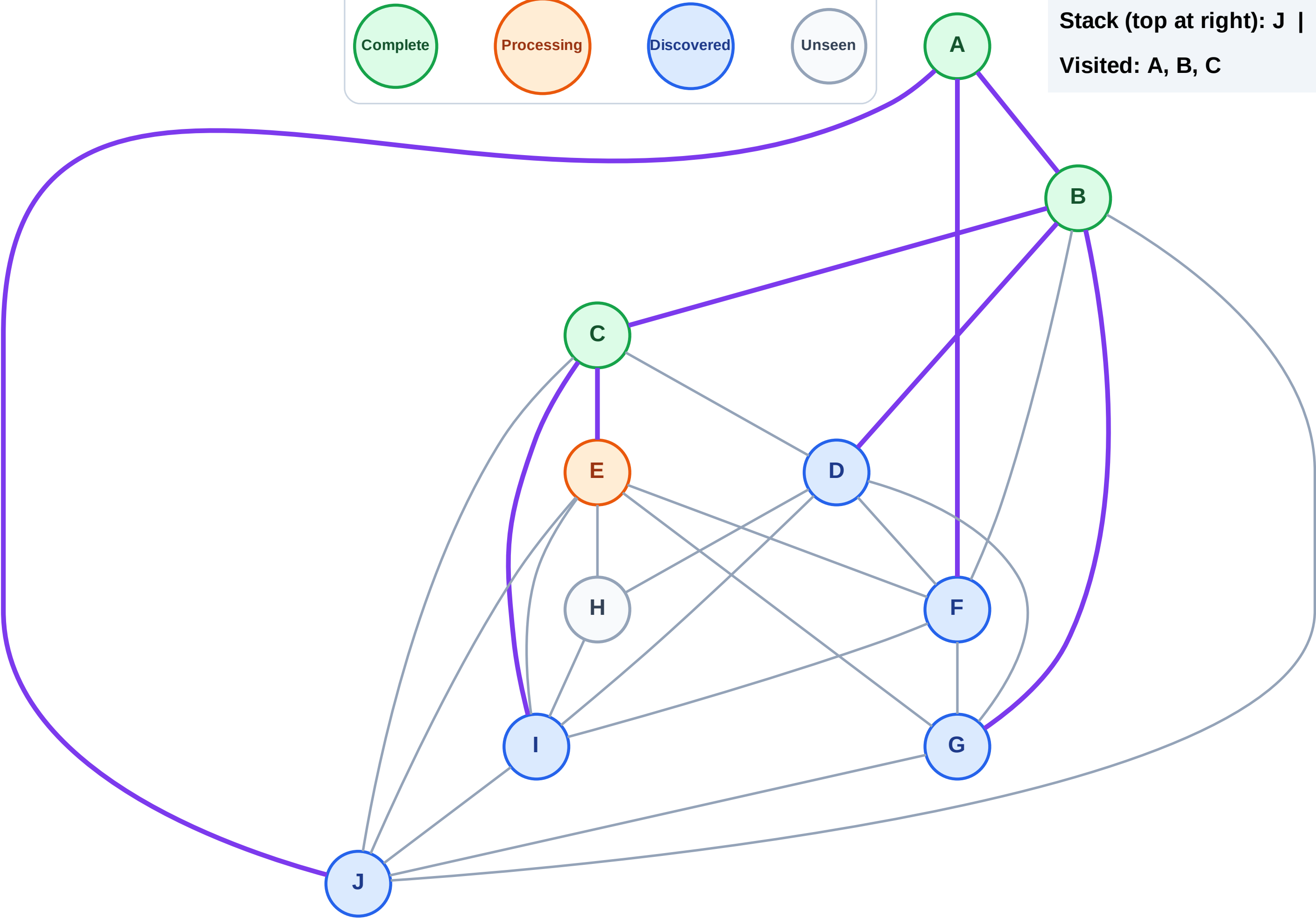
Processing

Discovered

Unseen

Stack (top at right): J | F | G | D | I

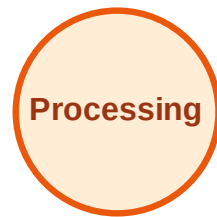
Visited: A, B, C



DFS Traversal

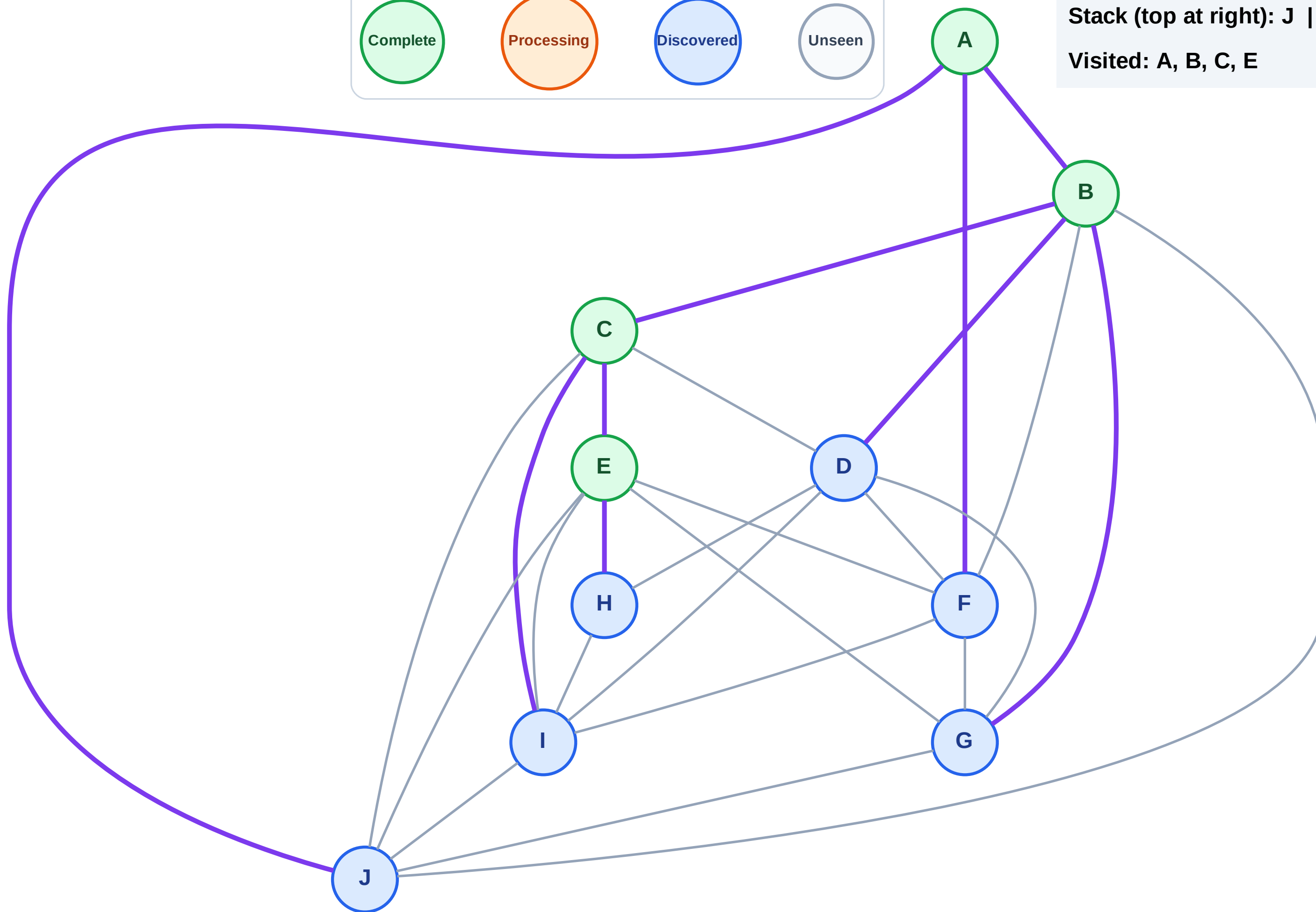
Step 9: Discover H from E

Legend



Stack (top at right): J | F | G | D | I | H

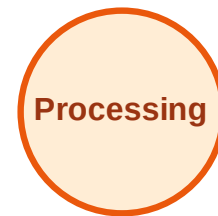
Visited: A, B, C, E



DFS Traversal

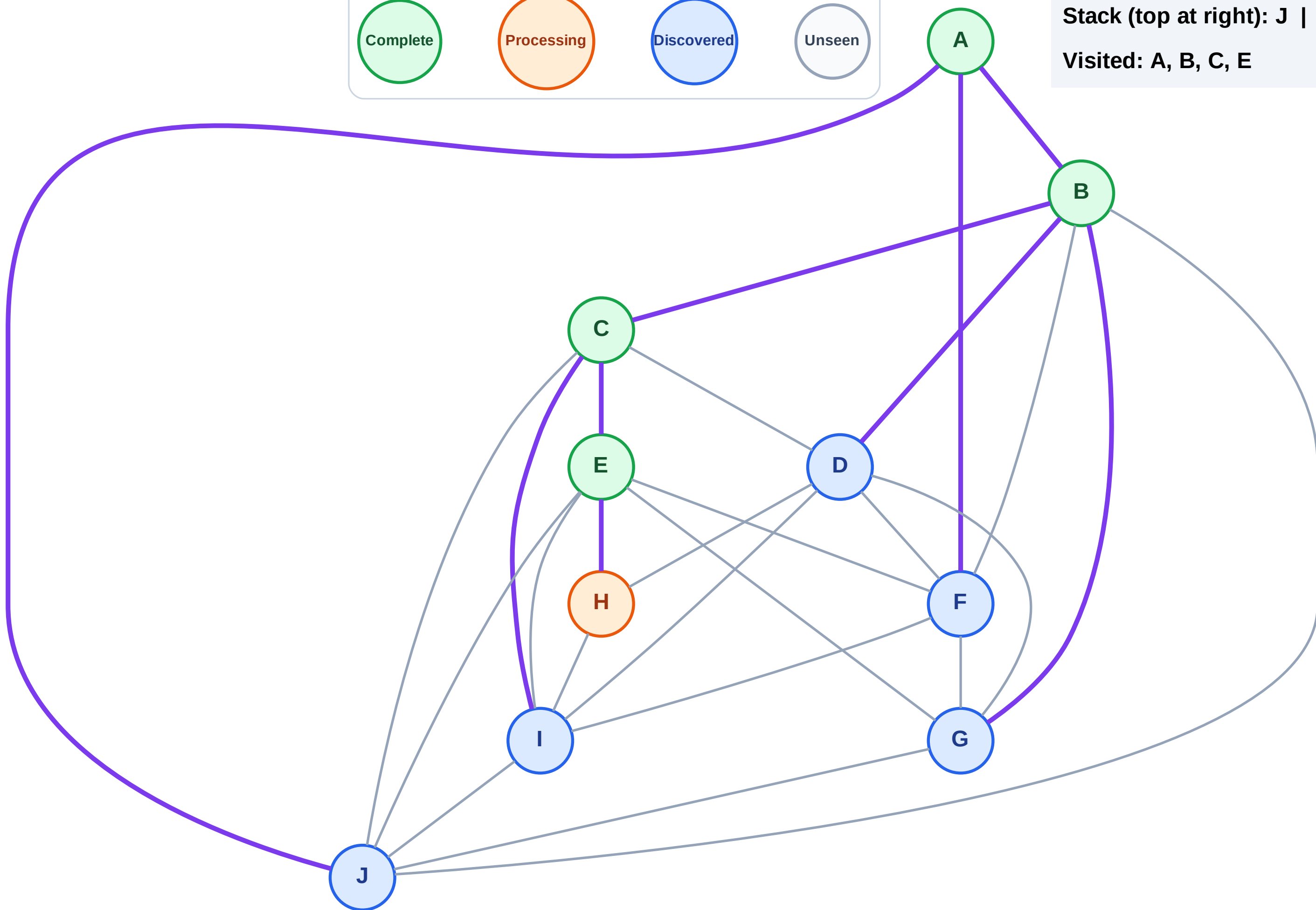
Step 10: Process node H

Legend



Stack (top at right): J | F | G | D | I

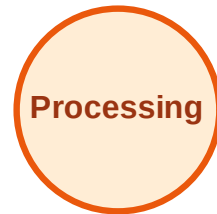
Visited: A, B, C, E



DFS Traversal

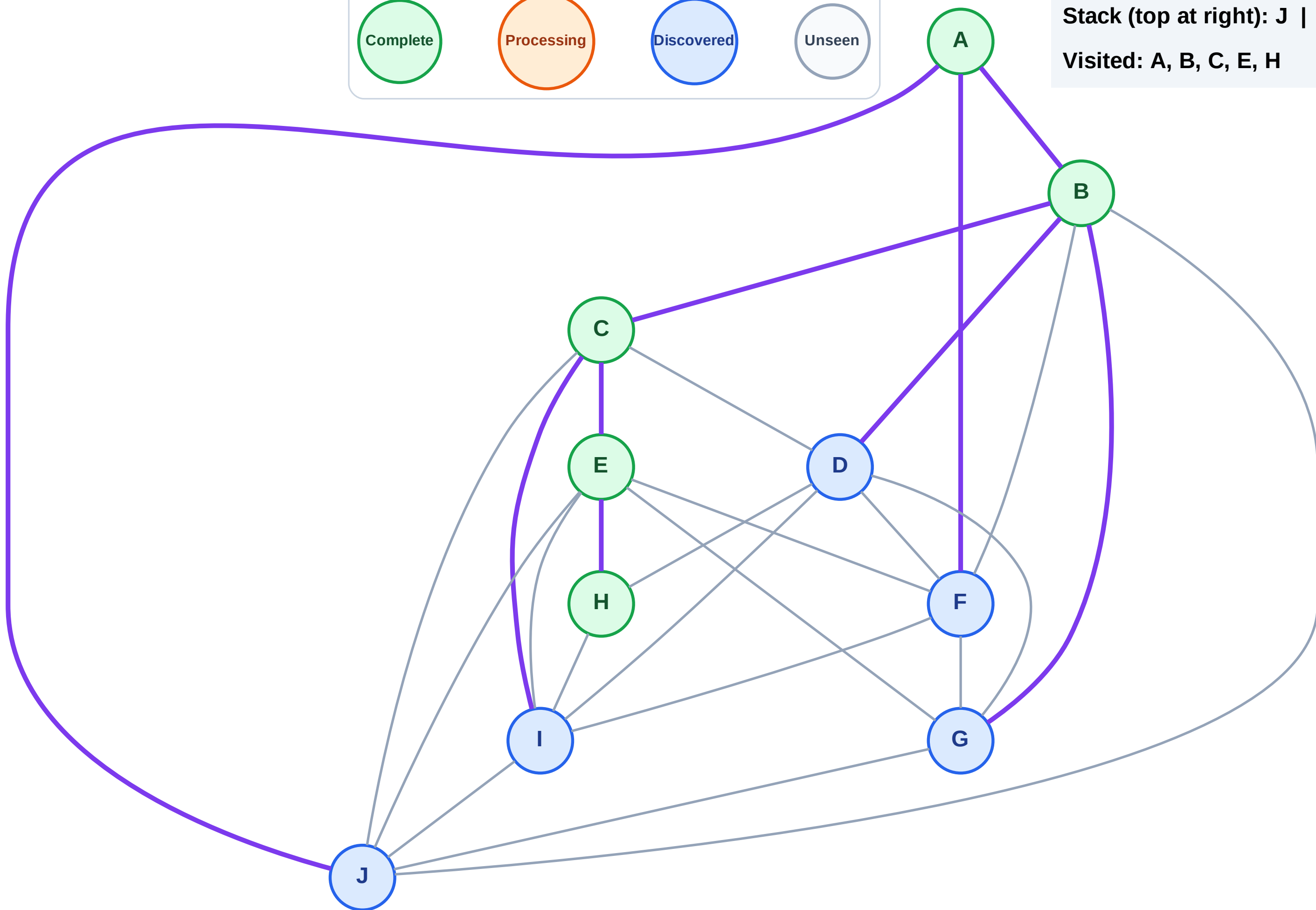
Step 11: No new neighbors from H

Legend



Stack (top at right): J | F | G | D | I

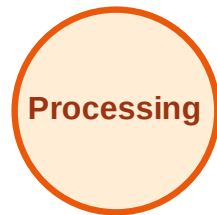
Visited: A, B, C, E, H



DFS Traversal

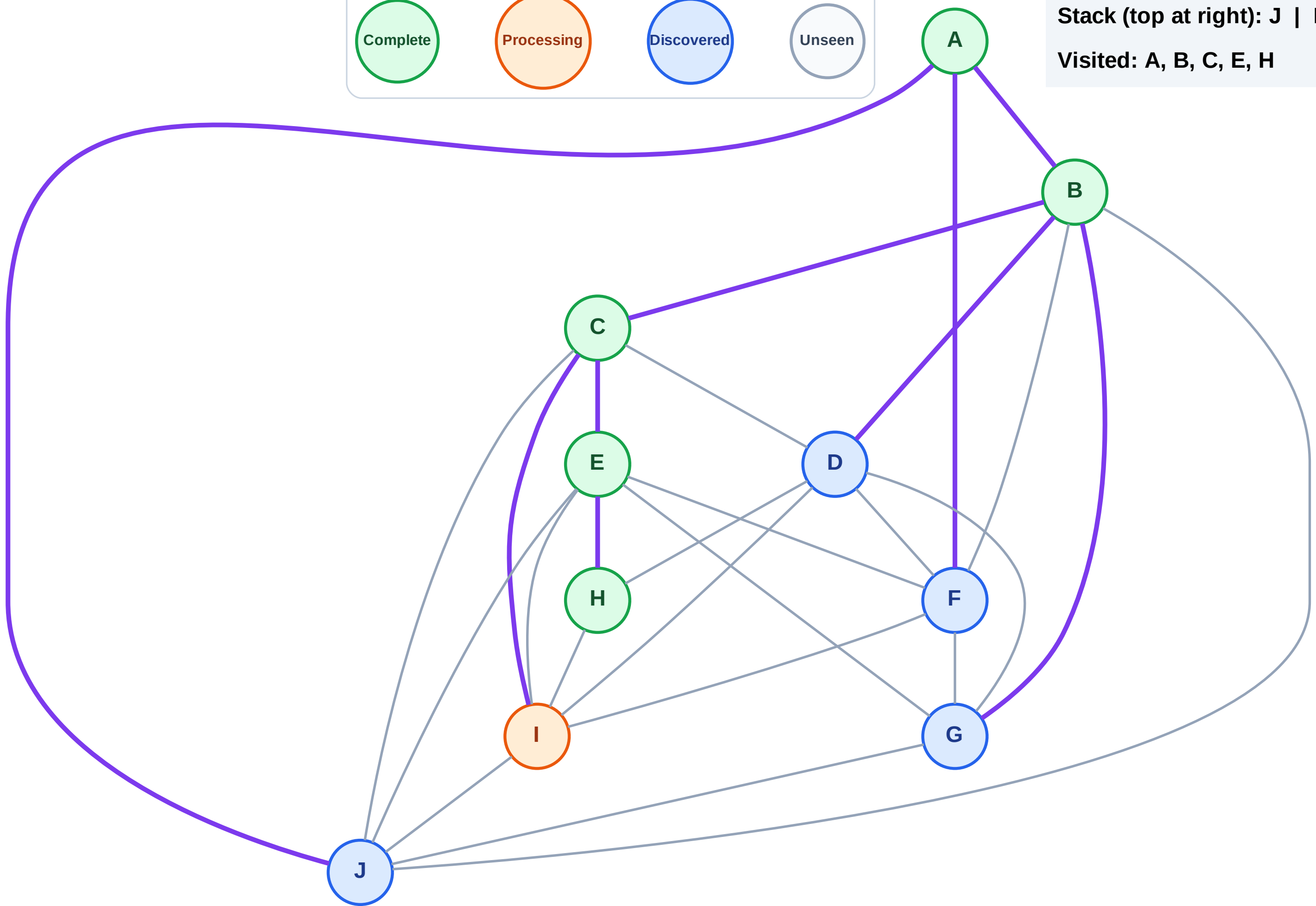
Step 12: Process node I

Legend



Stack (top at right): J | F | G | D

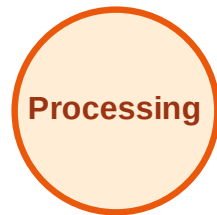
Visited: A, B, C, E, H



DFS Traversal

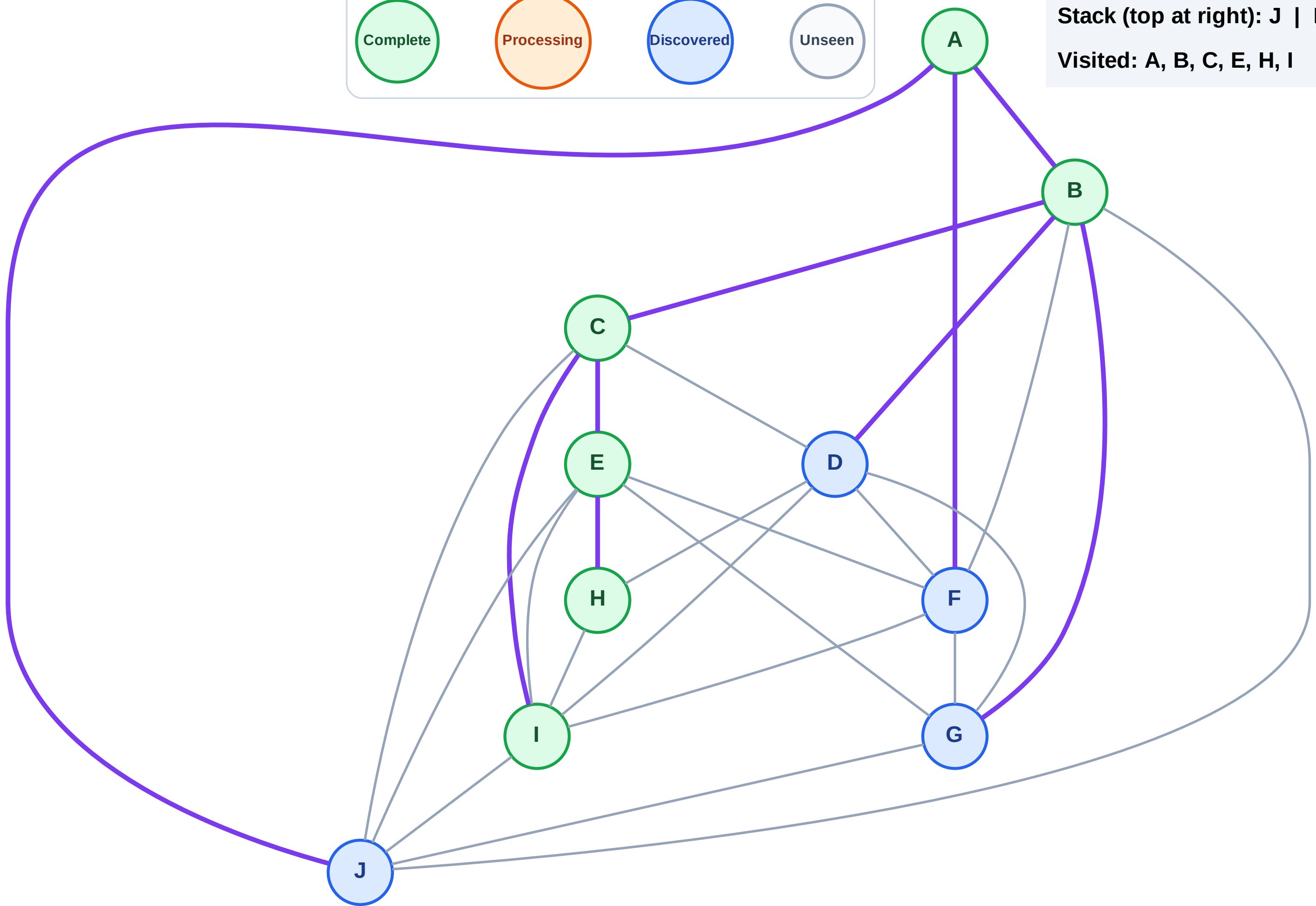
Step 13: No new neighbors from I

Legend



Stack (top at right): J | F | G | D

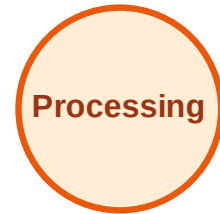
Visited: A, B, C, E, H, I



DFS Traversal

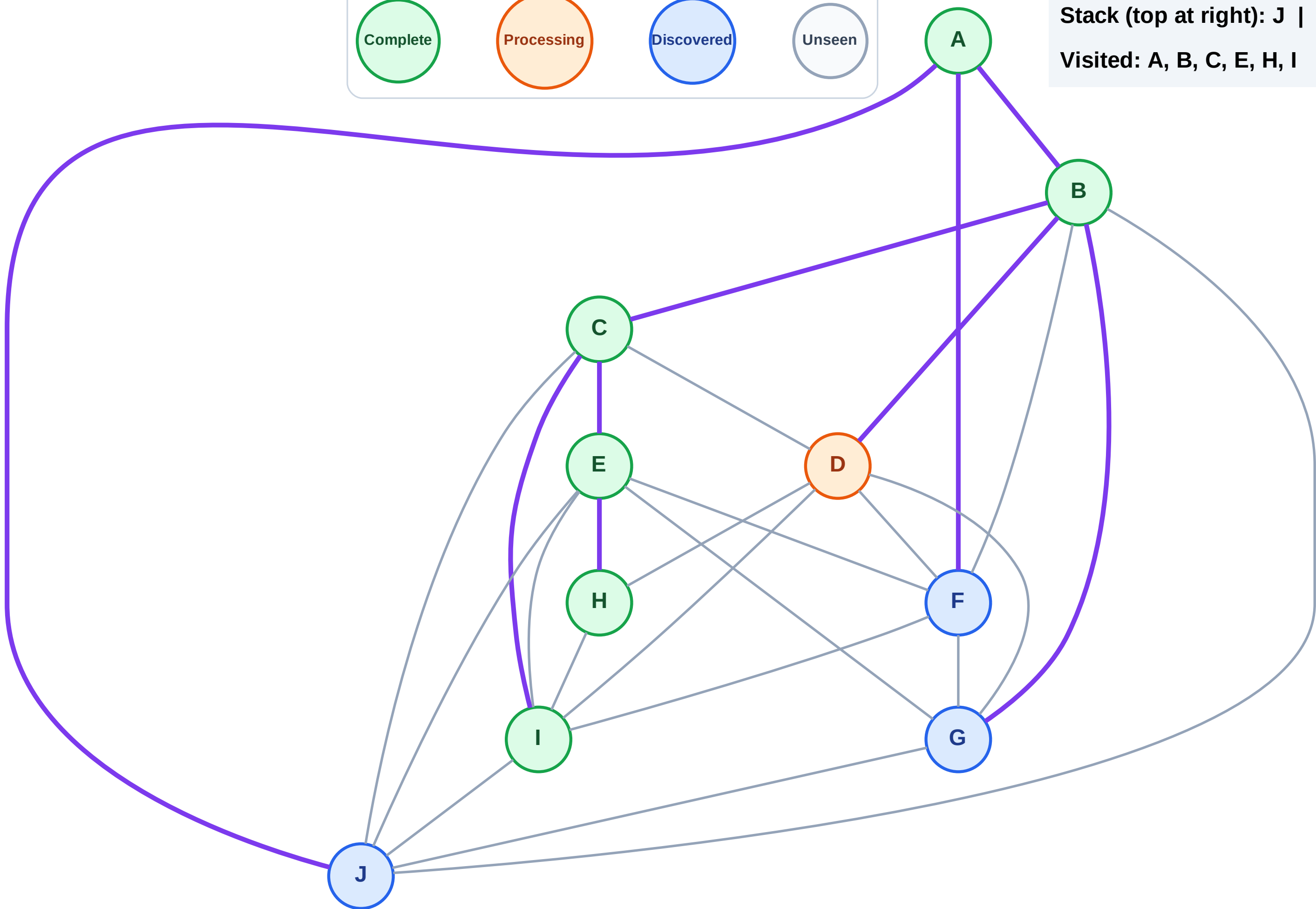
Step 14: Process node D

Legend



Stack (top at right): J | F | G

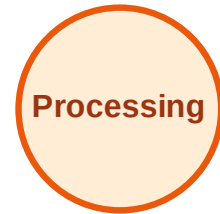
Visited: A, B, C, E, H, I



DFS Traversal

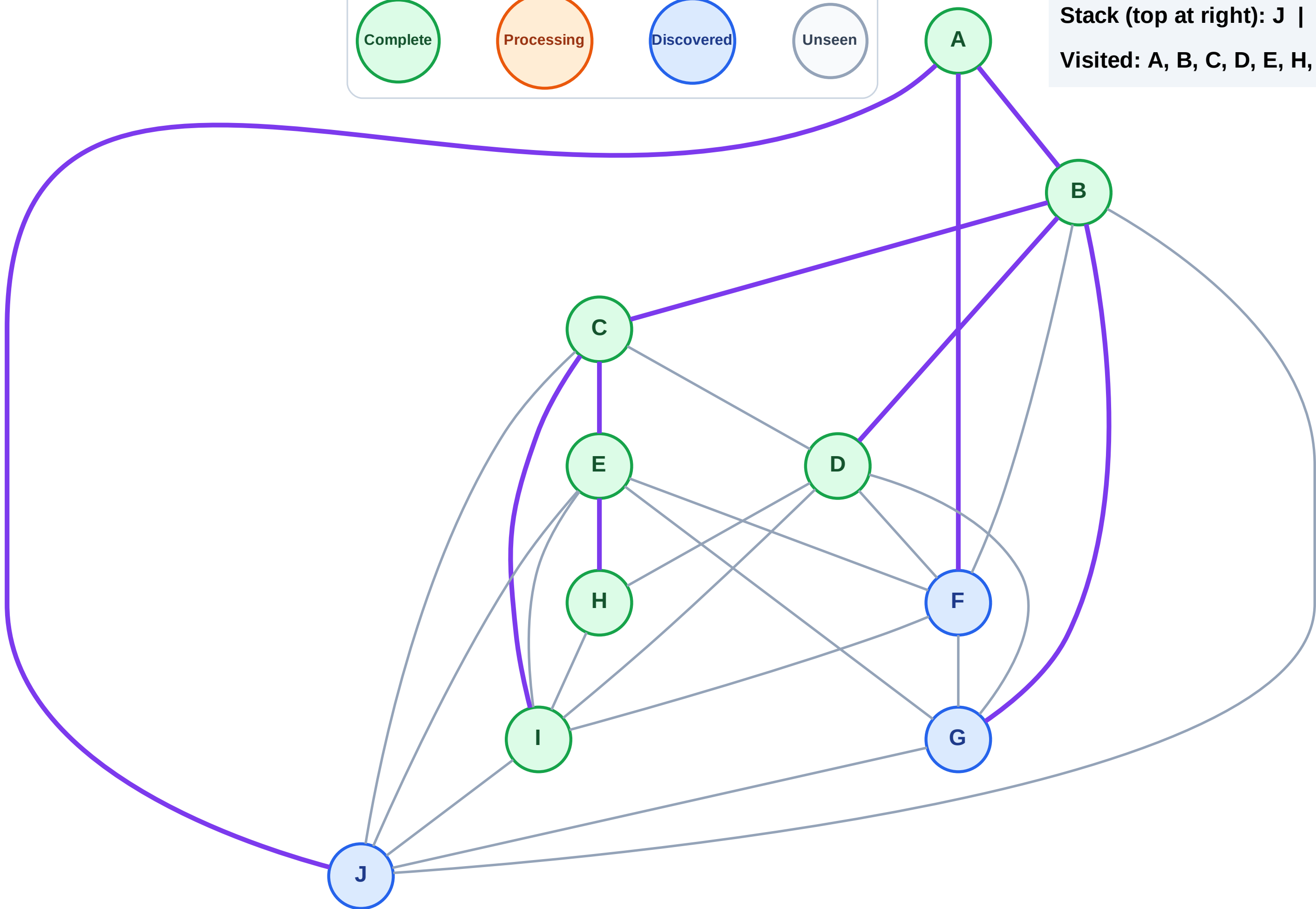
Step 15: No new neighbors from D

Legend



Stack (top at right): J | F | G

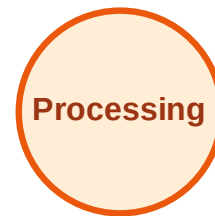
Visited: A, B, C, D, E, H, I



DFS Traversal

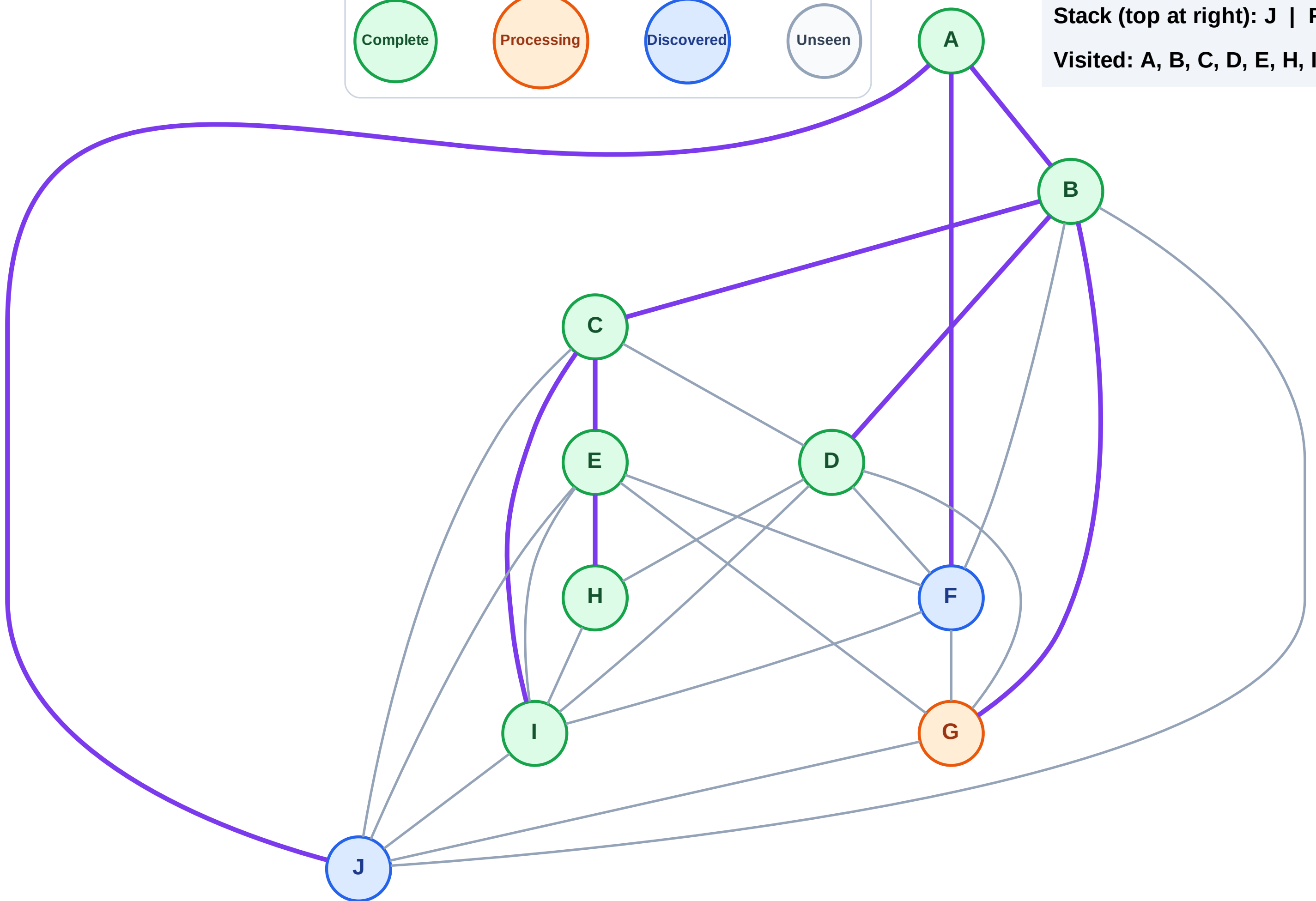
Step 16: Process node G

Legend



Stack (top at right): J | F

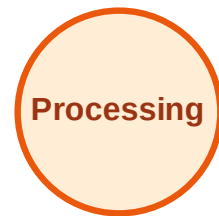
Visited: A, B, C, D, E, H, I



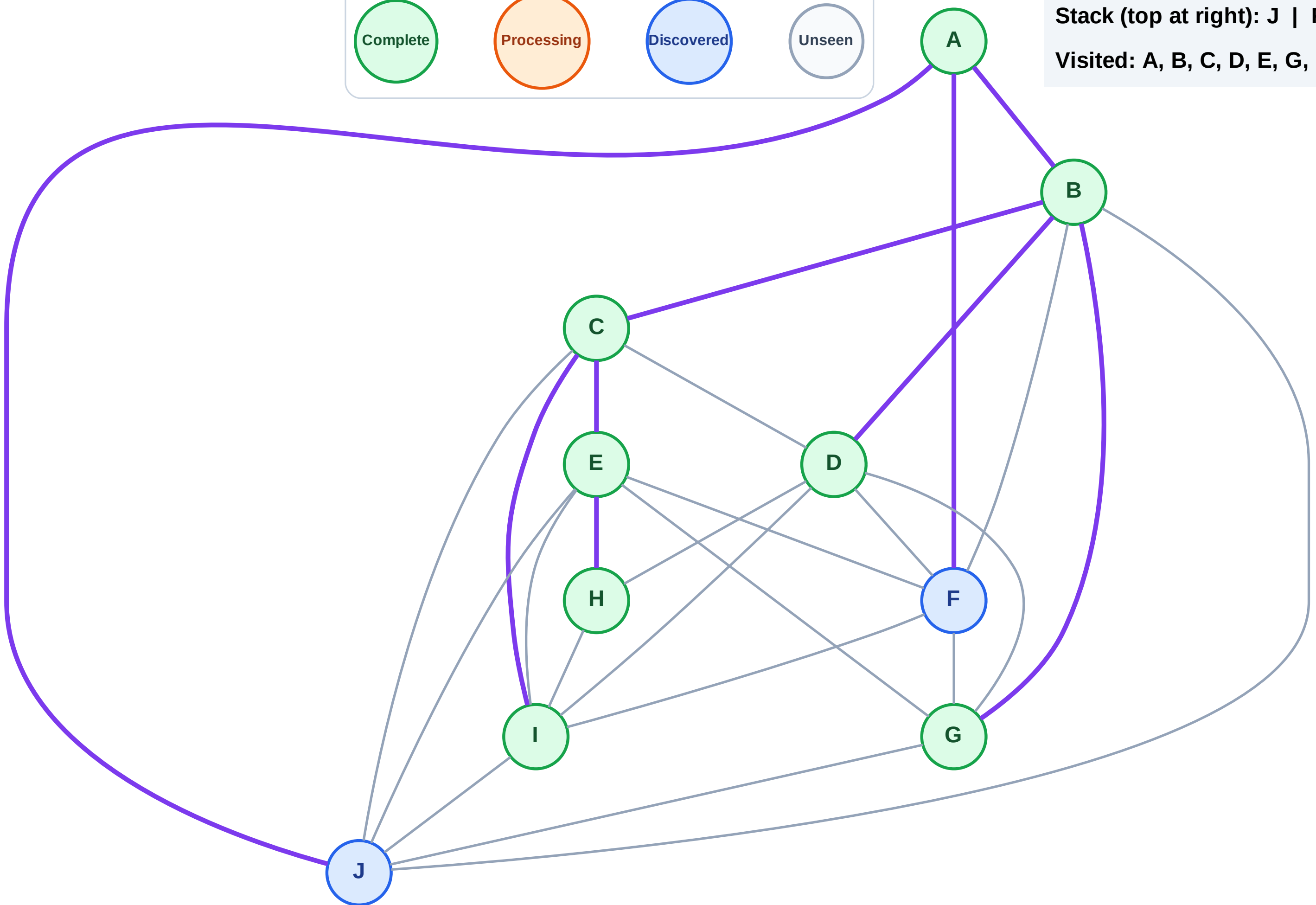
DFS Traversal

Step 17: No new neighbors from G

Legend



Stack (top at right): J | F
Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 18: Process node F

Legend

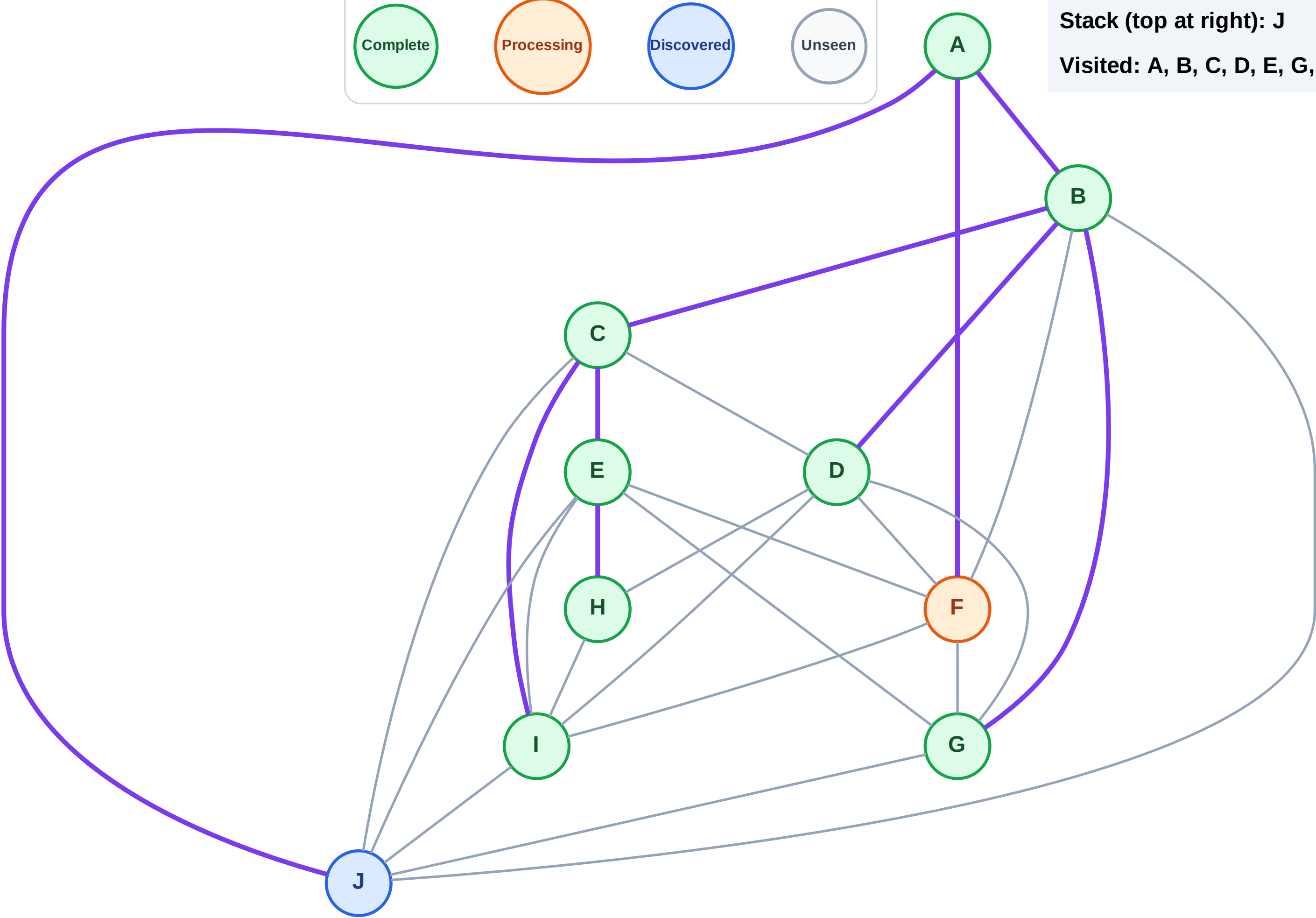
Complete

Processing

Discovered

Unseen

Stack (top at right): J
Visited: A, B, C, D, E, G, H, I



DFS Traversal

Step 19: No new neighbors from F

Legend

Complete

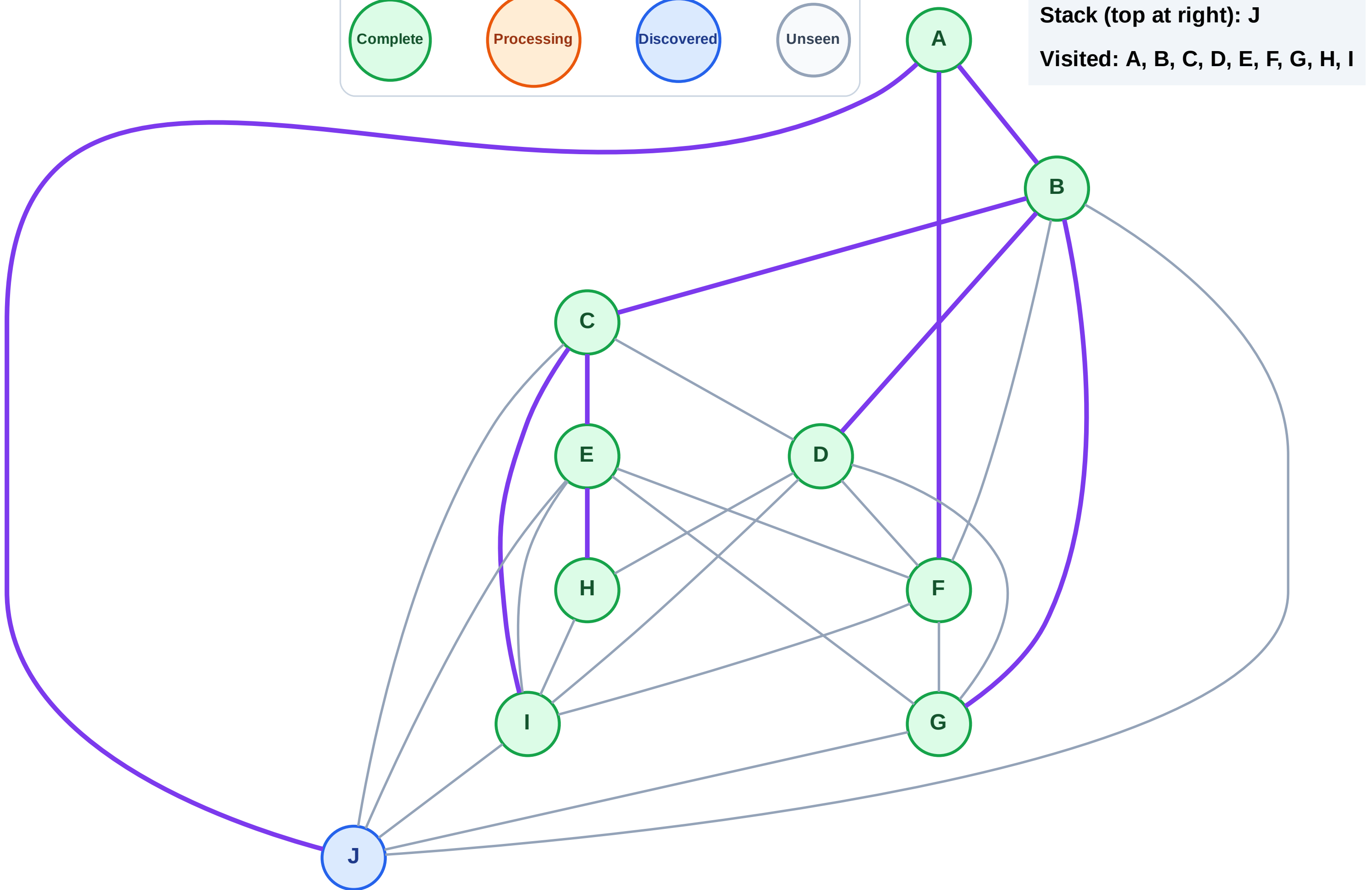
Processing

Discovered

Unseen

Stack (top at right): J

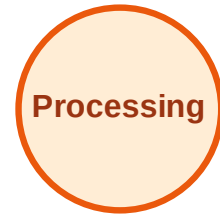
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

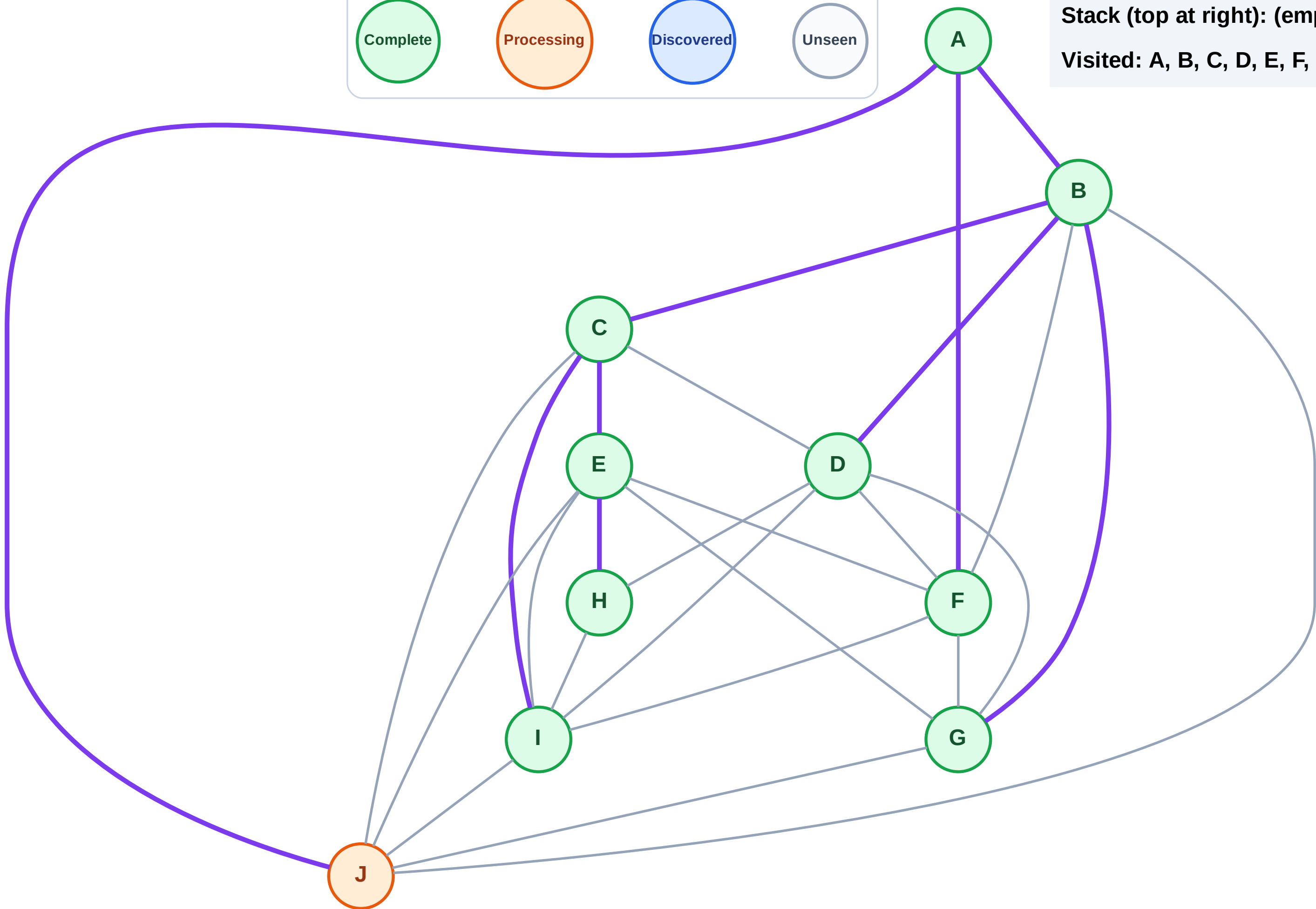
Step 20: Process node J

Legend



Stack (top at right): (empty)

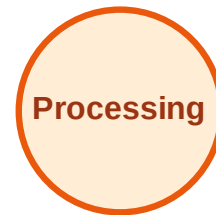
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

